מטלה מסכמת: מבוא ללמידת מכונה ומדעי הנתונים

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Imports

In [8]:
```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
from scipy.stats import pointbiserialr, chi2_contingency
import math
```

Helpfull Functions

In [9]:
```
def box_data_by_group(df, group_col, value_col):
    tmp = df[[group_col, value_col]].dropna(subset=[group_col, value_col])
    labels = sorted(tmp[group_col].unique())
    data = [tmp.loc[tmp[group_col] == g, value_col].values for g in labels]
    return labels, data
```

In [10]:
```
def crosstab_percent(df, row_col, col_col, dropna=True, round_to=2):
    data = df[[row_col, col_col]]

    if dropna:
        data = data.dropna(subset=[row_col, col_col])

    table = (
        pd.crosstab(data[row_col], data[col_col], normalize="index") * 100
    ).round(round_to)

    values = table.values
    row_labels = table.index.tolist()
    col_labels = table.columns.tolist()

    return table, values, row_labels, col_labels
```

In [11]:
```
def plot_heatmap(values, row_labels, col_labels, title="", xlabel="", ylabel="",
    """
    Draws a heatmap with numbers inside each cell.
    Designed to work with output from crosstab_percent.
    """
    plt.figure(figsize=figsize)
    plt.imshow(values, aspect="auto")

    plt.xticks(range(len(col_labels)), col_labels, rotation=45)
    plt.yticks(range(len(row_labels)), row_labels)
```

```
    plt.title(title)
    plt.xlabel(xlabel)
    plt.ylabel(ylabel)

    for i in range(len(row_labels)):
        for j in range(len(col_labels)):
            plt.text(j, i, format(values[i, j], fmt), ha="center", va="center",

    plt.colorbar(label="Percentage")
    plt.tight_layout()
    plt.show()
```

In [12]:
```
def plot_box_subplots(df, cols, group_col, nrows=2, ncols=2, figsize=(12,7), sho
    fig, axes = plt.subplots(nrows, ncols, figsize=figsize)

    if not isinstance(axes, (list, np.ndarray)):
        axes = [axes]
    else:
        axes = axes.ravel()

    for ax, col in zip(axes, cols):
        labels, data = box_data_by_group(df, group_col, col)
        ax.boxplot(data, labels=labels, showfliers=showfliers)
        ax.set_title(f"{col} by {group_col}")
        ax.set_xlabel(group_col)
        ax.set_ylabel(col)

    for ax in axes[len(cols):]:
        ax.axis("off")

    plt.tight_layout()
    plt.show()
```

In [13]:
```
def category_distribution(df, filter_col, filter_value, target_col, normalize=Tr
    """
    Returns percentage distribution of target_col
    within rows where filter_col == filter_value.
    """
    subset = df[df[filter_col] == filter_value]

    dist = (
        subset[target_col]
        .value_counts(normalize=normalize)
        .mul(100 if normalize else 1)
        .round(round_to)
    )

    return dist
```

In [14]:
```
def fill_missing_by_group_unique(df, group_col, target_col):
    def unique_value_or_nan(s):
        s = s.dropna()
        if len(s) == 0:
            return np.nan
        return s.iloc[0] if s.nunique() == 1 else np.nan

    group_map = df.groupby(group_col)[target_col].apply(unique_value_or_nan)
```

```
    missing_mask = df[target_col].isna()
    fill_mask = missing_mask & df[group_col].map(group_map).notna()

    df.loc[fill_mask, target_col] = df.loc[fill_mask, group_col].map(group_map)
```

In [15]:
```
def missing_table(df, col):
    return pd.DataFrame({
        "Count": [
            df[col].isna().sum(),
            df[col].notna().sum(),
            len(df)
        ]
    }, index=[
        f"{col} is NaN",
        f"{col} not NaN",
        "Total rows"
    ])
```

In [16]:
```
def scatter_plot(df, x_col, y_col, mask=None, title="", alpha=0.5, figsize=(8,5)
    if mask is not None:
        data = df.loc[mask, [x_col, y_col]]
    else:
        data = df[[x_col, y_col]].dropna()

    plt.figure(figsize=figsize)
    plt.scatter(data[x_col], data[y_col], alpha=alpha)
    plt.xlabel(x_col)
    plt.ylabel(y_col)
    plt.title(title)
    plt.tight_layout()
    plt.show()
```

חלק ראשון: אימון והערכת ביצועי מסווגים

טעינה ראשונית של המידע

In [17]:
```
spaceship_data = pd.read_csv('Spaceship_data.csv')
print(spaceship_data.shape)
```

```
(8693, 14)
```

הצגת מידע על הדאטה

In [18]:
```
spaceship_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8693 entries, 0 to 8692
Data columns (total 14 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   PassengerId   8693 non-null   object
 1   HomePlanet    8492 non-null   object
 2   CryoSleep     8476 non-null   object
 3   Cabin         8494 non-null   object
 4   Destination   8511 non-null   object
 5   Age           8514 non-null   float64
 6   VIP           8490 non-null   object
 7   RoomService   8512 non-null   float64
 8   FoodCourt     8510 non-null   float64
 9   ShoppingMall  8485 non-null   float64
 10  Spa           8510 non-null   float64
 11  VRDeck        8505 non-null   float64
 12  Name          8493 non-null   object
 13  Transported   8693 non-null   bool
dtypes: bool(1), float64(6), object(7)
memory usage: 891.5+ KB
```

בדיקת ייחודיות

In [19]:
```
spaceship_data.nunique()
```

Out[19]:
```
PassengerId     8693
HomePlanet         3
CryoSleep          2
Cabin           6560
Destination        3
Age               80
VIP                2
RoomService     1273
FoodCourt       1507
ShoppingMall    1115
Spa             1327
VRDeck          1306
Name            8473
Transported        2
dtype: int64
```

הצגת ראש המסמך

In [20]:
```
spaceship_data.head(10)
```

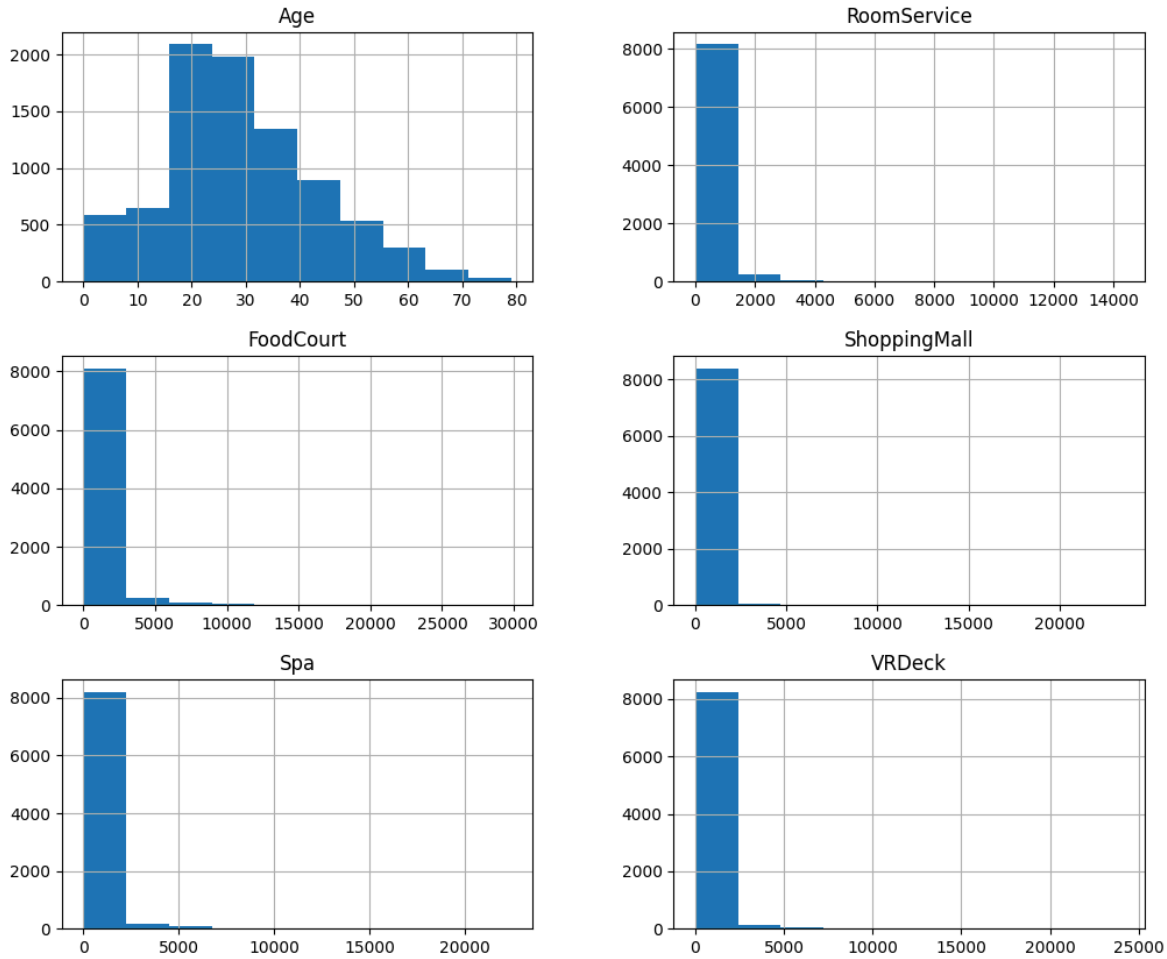
Out[20]:

	PassengerId	HomePlanet	CryoSleep	Cabin	Destination	Age	VIP	RoomService
0	0001_01	Europa	False	B/0/P	TRAPPIST-1e	39.0	False	0.0
1	0002_01	Earth	False	F/0/S	TRAPPIST-1e	24.0	False	109.0
2	0003_01	Europa	False	A/0/S	TRAPPIST-1e	58.0	True	43.0
3	0003_02	Europa	False	A/0/S	TRAPPIST-1e	33.0	False	0.0
4	0004_01	Earth	False	F/1/S	TRAPPIST-1e	16.0	False	303.0
5	0005_01	Earth	False	F/0/P	PSO J318.5-22	44.0	False	0.0
6	0006_01	Earth	False	F/2/S	TRAPPIST-1e	26.0	False	42.0
7	0006_02	Earth	True	G/0/S	TRAPPIST-1e	28.0	False	0.0
8	0007_01	Earth	False	F/3/S	TRAPPIST-1e	35.0	False	0.0
9	0008_01	Europa	True	B/1/P	55 Cancri e	14.0	False	0.0

הצגת היסטוגרמה

In [21]:
```
spaceship_data.hist(figsize=(12,10))
```

Out[21]:
```
array([[<Axes: title={'center': 'Age'}>,
        <Axes: title={'center': 'RoomService'}>],
       [<Axes: title={'center': 'FoodCourt'}>,
        <Axes: title={'center': 'ShoppingMall'}>],
       [<Axes: title={'center': 'Spa'}>,
        <Axes: title={'center': 'VRDeck'}>]], dtype=object)
```



הצגת התפלגויות מספריים

In [22]:
```
## sns.pairplot(spaceship_data)
```

חקר + שינוי טייפ הדאטה למספרים

In [23]:
```
spaceship_data_copy = spaceship_data.copy()
```

TotalSpend

In [24]:
```
spaceship_data_copy['TotalSpend'] = (
    spaceship_data_copy[['RoomService', 'FoodCourt', 'ShoppingMall', 'Spa', 'VRD
    .fillna(0)
    .sum(axis=1)
)
```

homeplanet

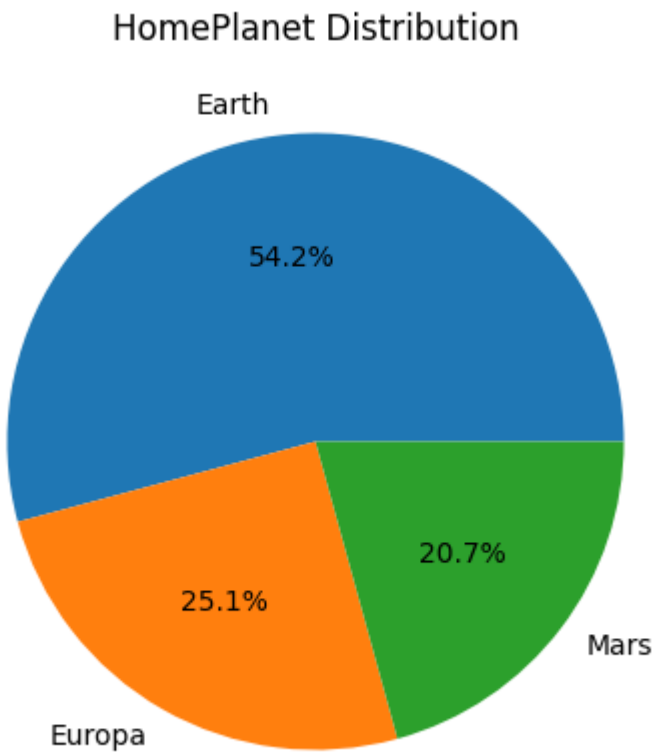
נתונים ראשוניים:

In [25]:
```
counts = spaceship_data_copy["HomePlanet"].value_counts()
print(counts)
```

```
HomePlanet
Earth     4602
Europa    2131
Mars      1759
Name: count, dtype: int64
```

In [26]:
```
counts.plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(5,5),
    title='HomePlanet Distribution',
    ylabel='',
)
```

Out[26]: <Axes: title={'center': 'HomePlanet Distribution'}>



מילון המרה משמות למספרים

In [27]:
```
homeplanet_map = {
    'Earth': 0,
    'Europa': 1,
    'Mars': 2
}
```

החלפת השמות בעזרת המילון

In [28]:
```
spaceship_data_copy['HomePlanet'] = (
    spaceship_data_copy['HomePlanet']
    .map(homeplanet_map)
    .fillna(-1)
    .astype(int)
)
```

destination

בדיקה כמה אופציות יש

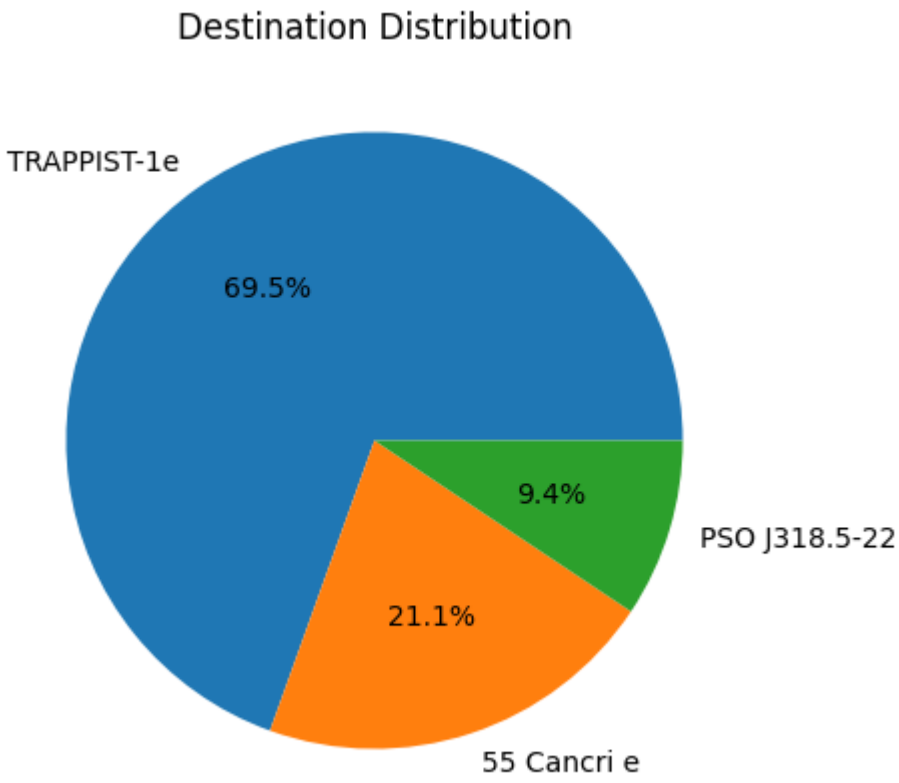
In [29]:
```
counts = spaceship_data_copy["Destination"].value_counts()
print(counts)
```

```
Destination
TRAPPIST-1e      5915
55 Cancri e      1800
PSO J318.5-22     796
Name: count, dtype: int64
```

הצגתם

In [30]:
```
counts.plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(5,5),
    title='Destination Distribution',
    ylabel='',
)
```

Out[30]: <Axes: title={'center': 'Destination Distribution'}>



יצירת מילון

In [31]:
```
destination_map = {
    'TRAPPIST-1e': 0,
    'PSO J318.5-22': 1,
    '55 Cancri e': 2
}
```

החלפת שמות במספרים

In [32]:
```
spaceship_data_copy['Destination'] = (
    spaceship_data_copy['Destination']
    .map(destination_map)
    .fillna(-1)
    .astype(int)
)
```

---

cryosleep

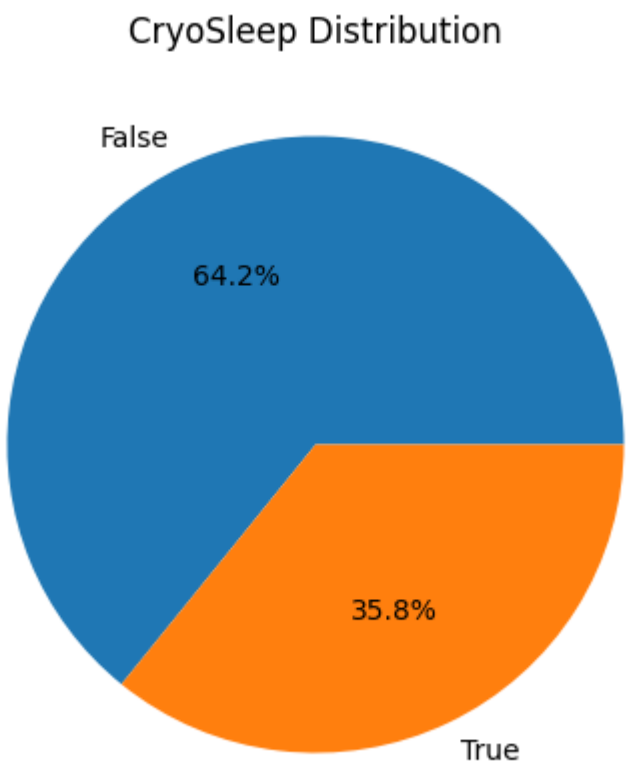
התפלגות

In [33]:
```
counts = spaceship_data_copy["CryoSleep"].value_counts()
print(counts)
```

```
CryoSleep
False    5439
True     3037
Name: count, dtype: int64
```

In [34]:
```
counts.plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(5,5),
    title='CryoSleep Distribution',
    ylabel='',
)
```

Out[34]:
```
<Axes: title={'center': 'CryoSleep Distribution'}>
```



המרה ממשתנה בוליאני לאינטגרים

In [35]:
```
spaceship_data_copy['CryoSleep'] = spaceship_data_copy['CryoSleep'].map({True: 1
```

VIP

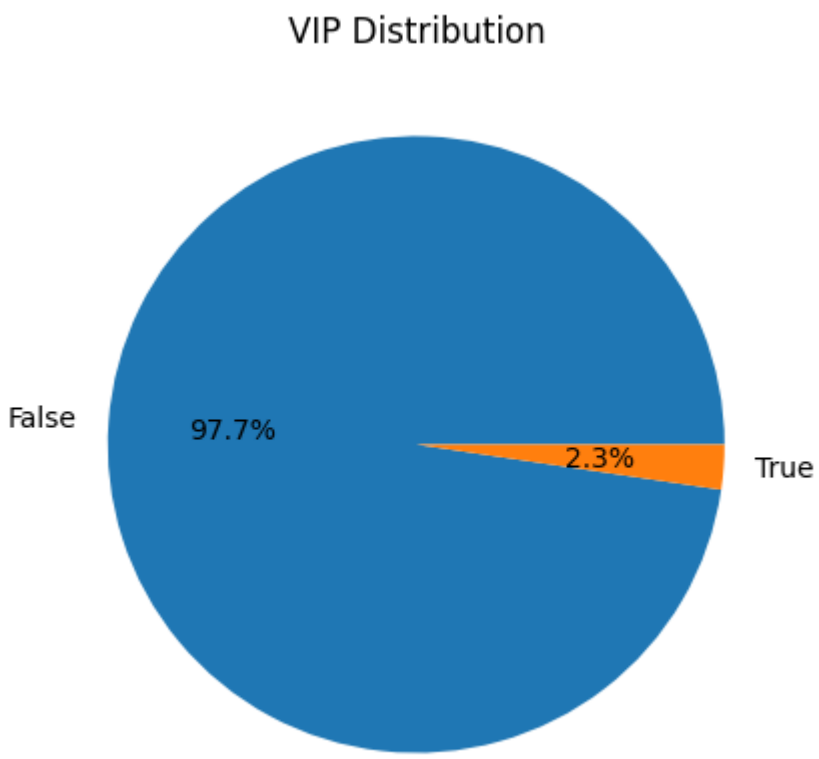
התפלגות

In [36]:
```
counts = spaceship_data_copy["VIP"].value_counts()
print(counts)
```

```
VIP
False    8291
True      199
Name: count, dtype: int64
```

In [37]:
```
counts.plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(5,5),
    title='VIP Distribution',
    ylabel='',
)
```

Out[37]:
```
<Axes: title={'center': 'VIP Distribution'}>
```



החלפת משתנה בוליאני במשתנה מספרי

In [38]:
```
spaceship_data_copy['VIP'] = spaceship_data_copy['VIP'].map({True: 1, False: 0})
```

Transported

שינוי משתנה בוליאני במשתנה מספרי

In [39]:
```
spaceship_data_copy['Transported'] = spaceship_data_copy['Transported'].map({Tru
```

PassengerID

בדיקת ייחודיות שלא יש כפילויות

In [40]:
```
id_check = spaceship_data_copy['PassengerId'].value_counts()
print(id_check[id_check > 1])
```

```
Series([], Name: count, dtype: int64)
```

בדיקת תקינות שהמילון לא יתקע

In [41]:
```
pattern = r'^\d{4}_\d{2}$'
invalid_ids = spaceship_data_copy[~spaceship_data_copy['PassengerId'].str.match(
print(invalid_ids)
```

```
Empty DataFrame
Columns: [PassengerId, HomePlanet, CryoSleep, Cabin, Destination, Age, VIP, RoomS
ervice, FoodCourt, ShoppingMall, Spa, VRDeck, Name, Transported, TotalSpend]
Index: []
```

פיצול העמודה ל2 עמודות:

- PassengerId -> GroupId, GroupMemberId

In [42]:
```
spaceship_data_copy['GroupId'] = spaceship_data_copy['PassengerId'].str.split('_
spaceship_data_copy['GroupMemberId'] = spaceship_data_copy['PassengerId'].str.sp

spaceship_data_copy['GroupId'] = spaceship_data_copy['GroupId'].astype(int)
spaceship_data_copy['GroupMemberId'] = spaceship_data_copy['GroupMemberId'].asty

spaceship_data_copy.drop(columns=['PassengerId'], inplace=True)
```

שינוי Cabin

בדיקת תקינות

In [43]:
```
pattern =  r'^[A-GT]/\d+/(P|S)$'
is_valid = (
    spaceship_data_copy['Cabin'].isna() |
    spaceship_data_copy['Cabin'].str.match(pattern)
)
```

In [44]:
```
validity_summary = (
    is_valid
    .value_counts()
    .rename({True: 'Valid (or NaN)', False: 'Invalid'})
    .reset_index()
)

validity_summary.columns = ['Cabin_validity', 'Count']
validity_summary
```

Out[44]:

	Cabin_validity	Count
0	Valid (or NaN)	8693

פיצול הCABIN למידע:

- Cabin -> Deck,RoomNum,RoomSide

In [45]:
```
spaceship_data_copy['Deck'] = spaceship_data['Cabin'].str.split('/').str[0]
spaceship_data_copy['RoomNum'] = spaceship_data['Cabin'].str.split('/').str[1]
spaceship_data_copy['RoomSide'] = spaceship_data['Cabin'].str.split('/').str[2]

spaceship_data_copy['RoomNum'] = spaceship_data_copy['RoomNum'].astype('float64'
```

In [46]:
```
deck_map = {
    'A': 0,
    'B': 1,
```

```
    'C': 2,
    'D': 3,
    'E': 4,
    'F': 5,
    'G': 6,
    'T': 7
}

roomside_map = {
    'P': 0,
    'S': 1,
}
```

In [47]:
```
spaceship_data_copy['Deck'] = (
    spaceship_data_copy['Deck']
    .map(deck_map)
    .fillna(-1)
    .astype(int)
)

spaceship_data_copy['RoomSide'] = (
    spaceship_data_copy['RoomSide']
    .map(roomside_map)
    .fillna(-1)
    .astype(int)
)
```

Name

פיצול השם לשם פרטי ושם משפחה

In [48]:
```
spaceship_data_copy[['FirstName', 'Surname']] = spaceship_data['Name'].str.split
```

בדיקת שלכולם יש 2 שמות

In [49]:
```
name_vector = spaceship_data['Name']

is_valid_name = (
    name_vector.isna() |                        # NaN תקין
    name_vector.str.match(r'^[^ ]+ [^ ]+$', na=False)
)
```

In [50]:
```
name_validity_table = (
    is_valid_name
    .value_counts()
    .rename({True: 'Valid (two words)', False: 'Invalid'})
    .to_frame(name='Count')
)

name_validity_table
```

Out[50]:

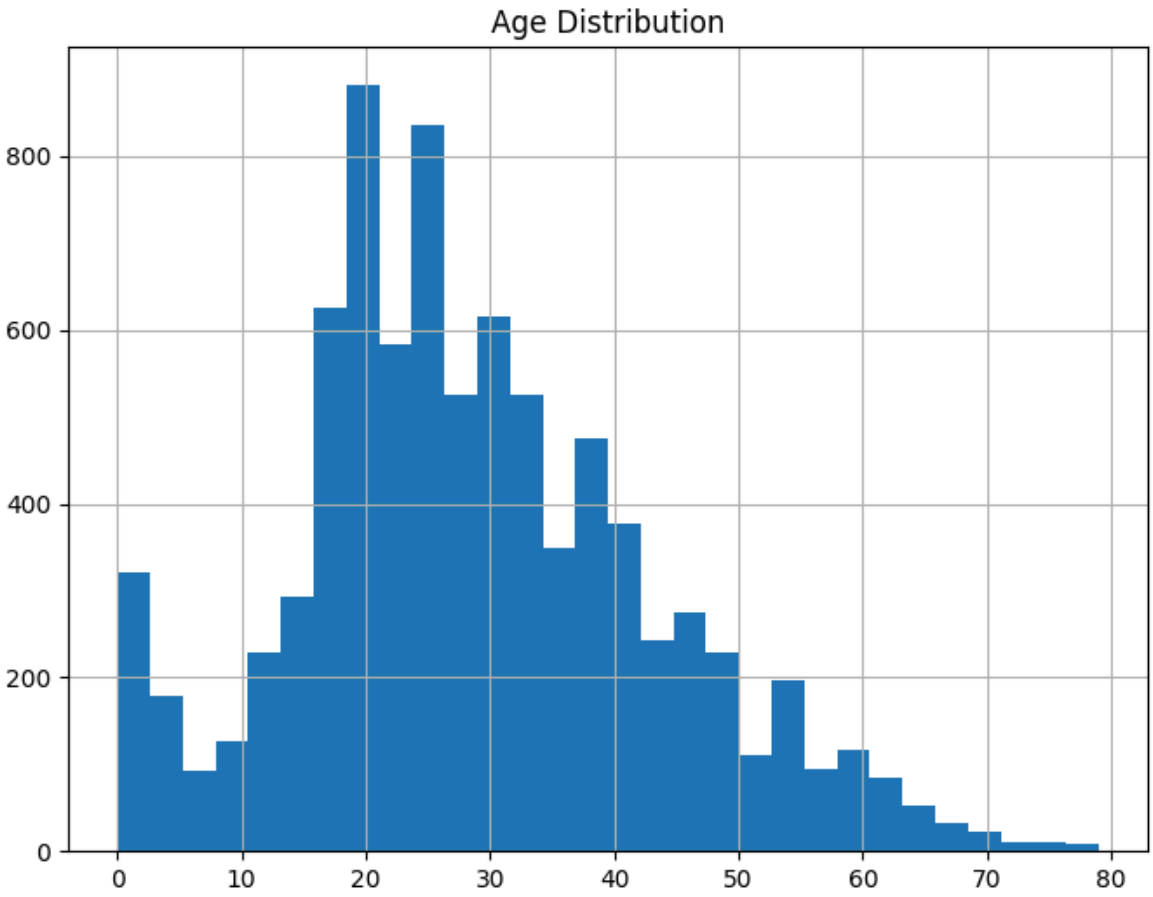
Name	Count
Valid (two words)	8693

Age

התפלגות גילאים

In [51]:
```
ax = spaceship_data_copy['Age'].hist(figsize=(8,6), bins=30)
ax.set_title('Age Distribution')
```

Out[51]: Text(0.5, 1.0, 'Age Distribution')

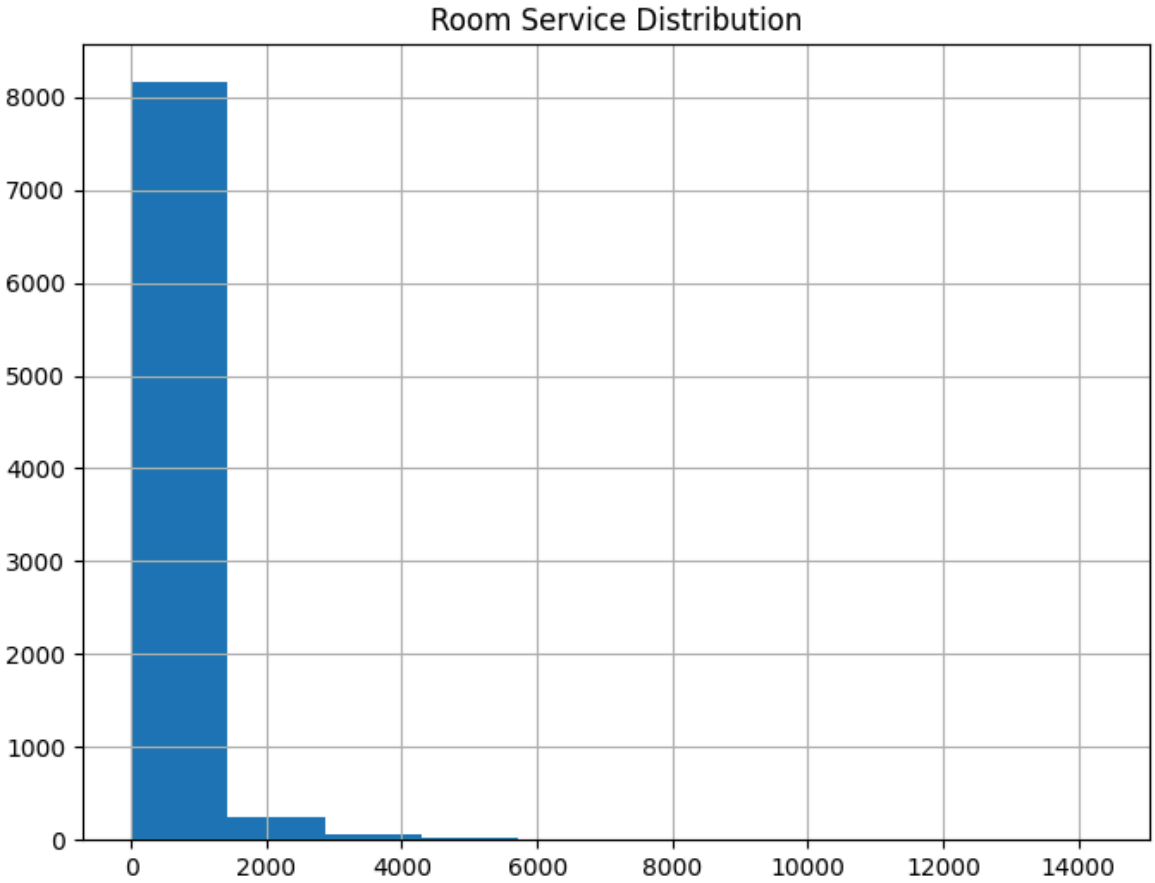


RoomService

התפלגות

In [52]:
```
ax = spaceship_data_copy['RoomService'].hist(figsize=(8,6))
ax.set_title('Room Service Distribution')
```

Out[52]: Text(0.5, 1.0, 'Room Service Distribution')

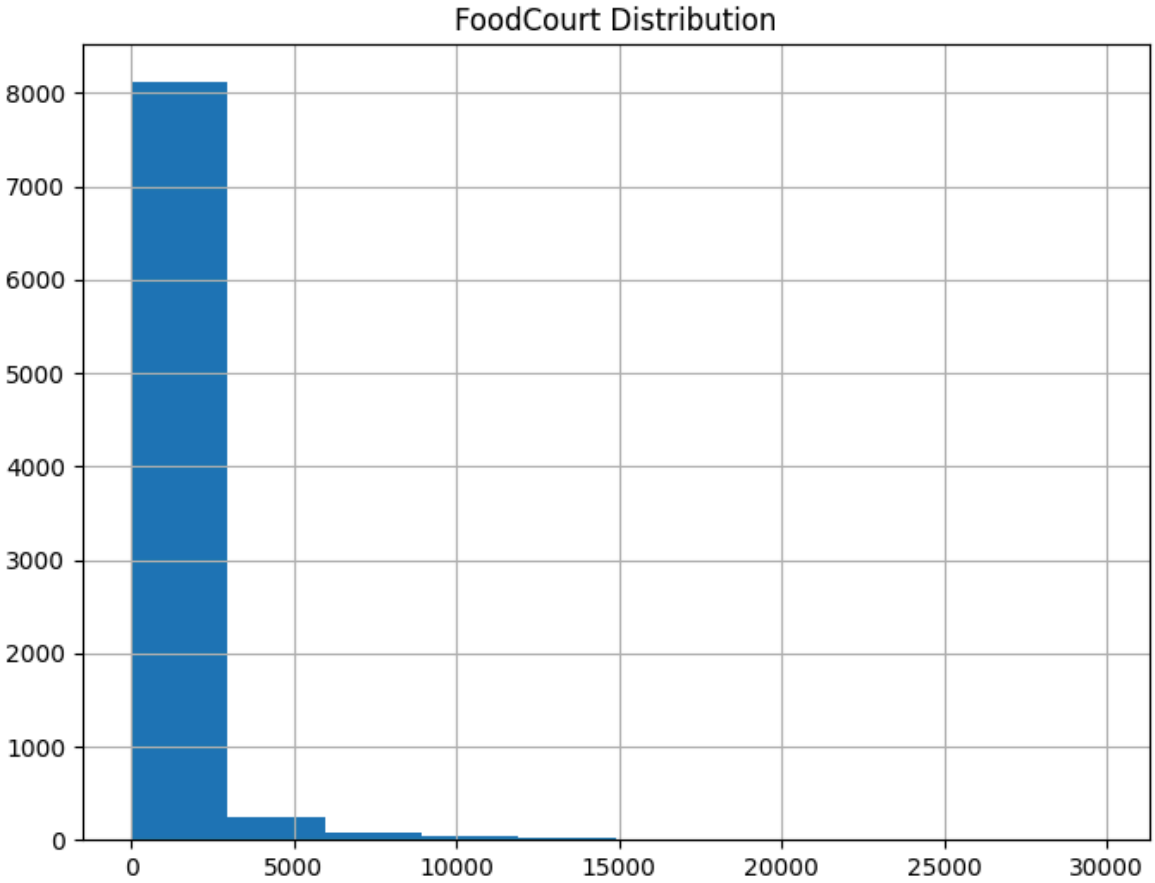


FoodCourt

התפלגות

In [53]:
```
ax = spaceship_data_copy['FoodCourt'].hist(figsize=(8,6))
ax.set_title('FoodCourt Distribution')
```

Out[53]:
```
Text(0.5, 1.0, 'FoodCourt Distribution')
```

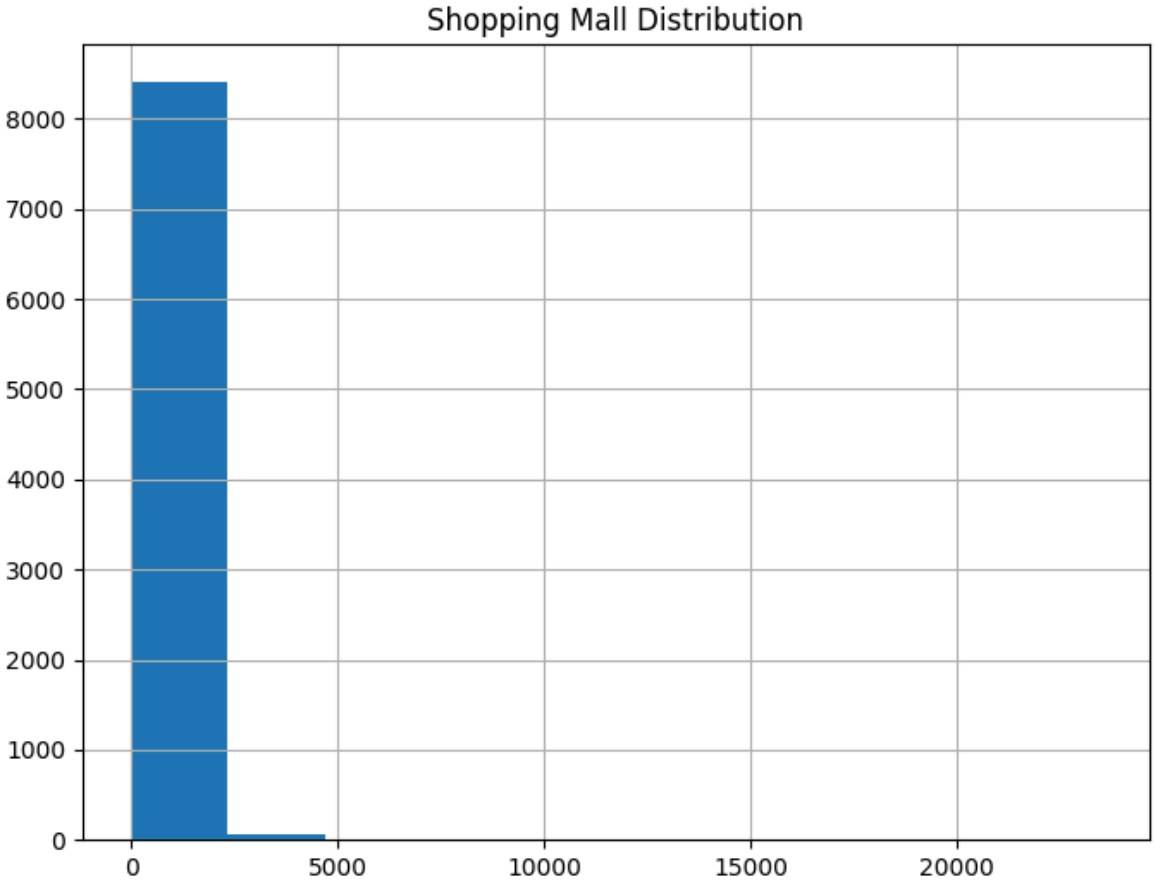


ShoppingMall

התפלגות

In [54]:
```
ax = spaceship_data_copy['ShoppingMall'].hist(figsize=(8,6))
ax.set_title('Shopping Mall Distribution')
```

Out[54]:
```
Text(0.5, 1.0, 'Shopping Mall Distribution')
```

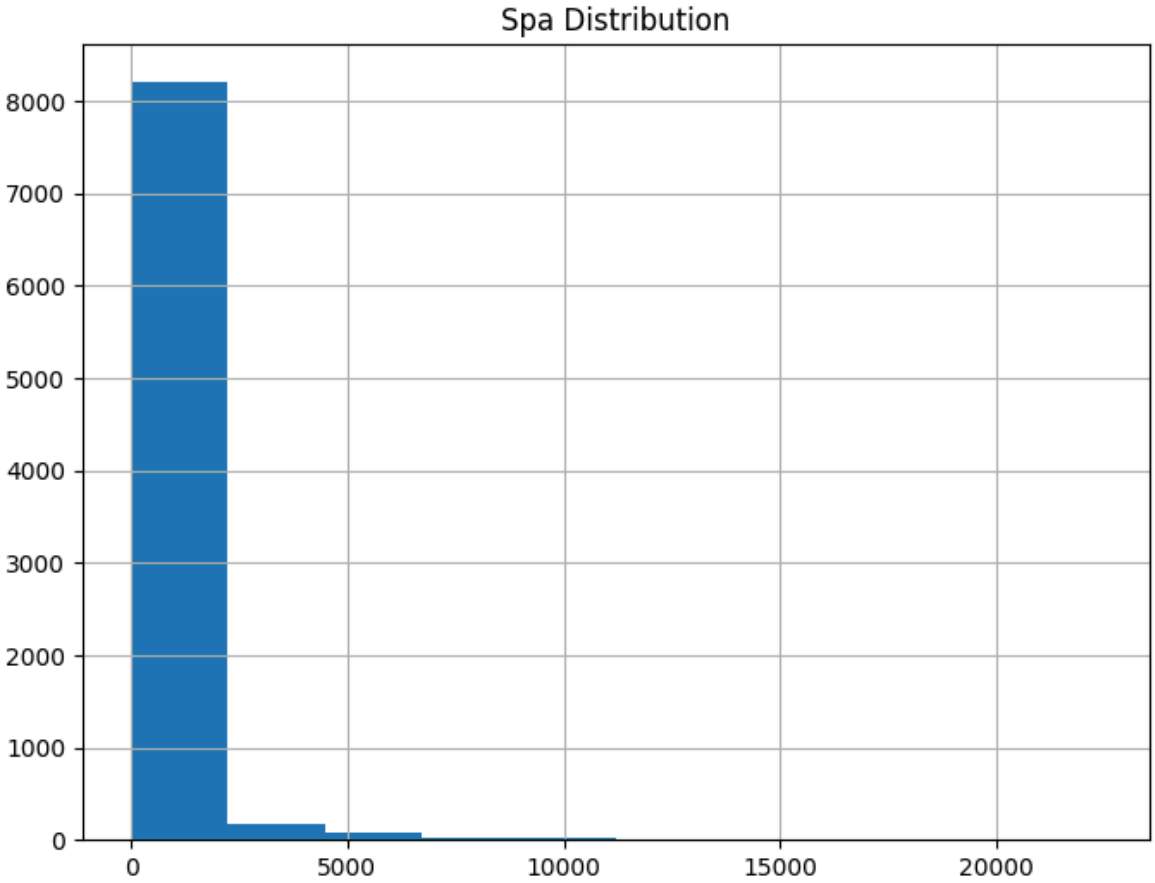


Spa

התפלגות

In [55]:
```
ax = spaceship_data_copy['Spa'].hist(figsize=(8,6))
ax.set_title('Spa Distribution')
```

Out[55]:
```
Text(0.5, 1.0, 'Spa Distribution')
```

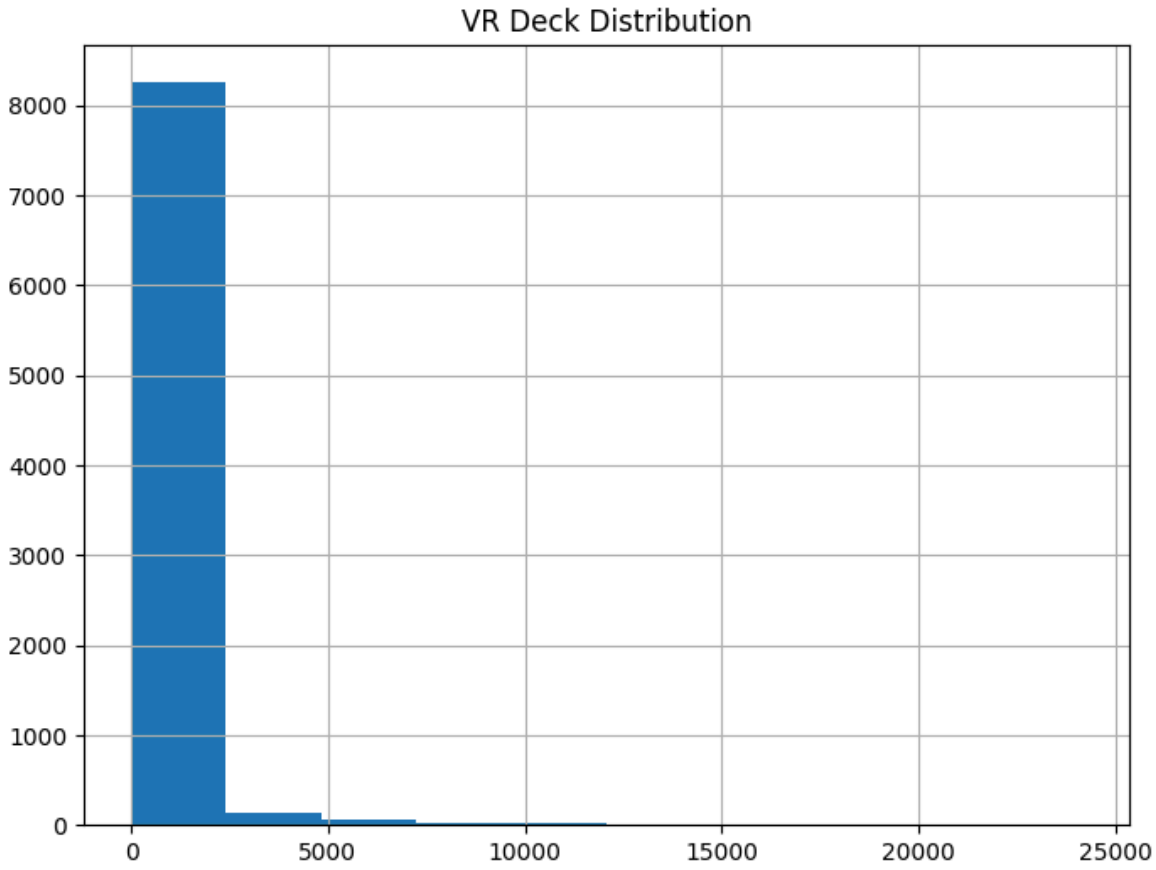


VRDeck

התפלגות

In [56]:
```
ax = spaceship_data_copy['VRDeck'].hist(figsize=(8,6))
ax.set_title('VR Deck Distribution')
```

Out[56]:
```
Text(0.5, 1.0, 'VR Deck Distribution')
```



החזרת מידע חסר

In [57]:
```
spaceship_data = spaceship_data_copy.copy()
spaceship_data.replace(-1, np.nan, inplace=True)
spaceship_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8693 entries, 0 to 8692
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   HomePlanet     8492 non-null   float64
 1   CryoSleep      8476 non-null   float64
 2   Cabin          8494 non-null   object
 3   Destination    8511 non-null   float64
 4   Age            8514 non-null   float64
 5   VIP            8490 non-null   float64
 6   RoomService    8512 non-null   float64
 7   FoodCourt      8510 non-null   float64
 8   ShoppingMall   8485 non-null   float64
 9   Spa            8510 non-null   float64
 10  VRDeck         8505 non-null   float64
 11  Name           8493 non-null   object
 12  Transported    8693 non-null   int64
 13  TotalSpend     8693 non-null   float64
 14  GroupId        8693 non-null   int64
 15  GroupMemberId  8693 non-null   int64
 16  Deck           8494 non-null   float64
 17  RoomNum        8494 non-null   float64
 18  RoomSide       8494 non-null   float64
 19  FirstName      8493 non-null   object
 20  Surname        8493 non-null   object
dtypes: float64(14), int64(3), object(4)
memory usage: 1.4+ MB
```

מציאת ערכים חסרים בדאטה

HomePlanet

homeplanet_map = { 'Earth': 0, 'Europa': 1, 'Mars': 2 }

In [58]:
```
missing_table(spaceship_data, "HomePlanet")
```

Out[58]:

	Count
HomePlanet is NaN	201
HomePlanet not NaN	8492
Total rows	8693

הצגת המידע בחיתוכים עם יתר הדאטה:

בדיקת הוצאת כספים לפי כוכב אם

In [59]:
```
money_cols = ["RoomService", "FoodCourt", "ShoppingMall", "Spa", "VRDeck"]
money_cols = [c for c in money_cols if c in spaceship_data.columns]
```

In [60]:
```
money_summary_table = (
    spaceship_data
    .dropna(subset=["HomePlanet"])
    .groupby("HomePlanet")[money_cols]
```

```
    .agg(["count", "mean", "median"])
    .round(2)
)

money_summary_table.columns = [f"{col}_{stat}" for col, stat in money_summary_ta
money_summary_table
```

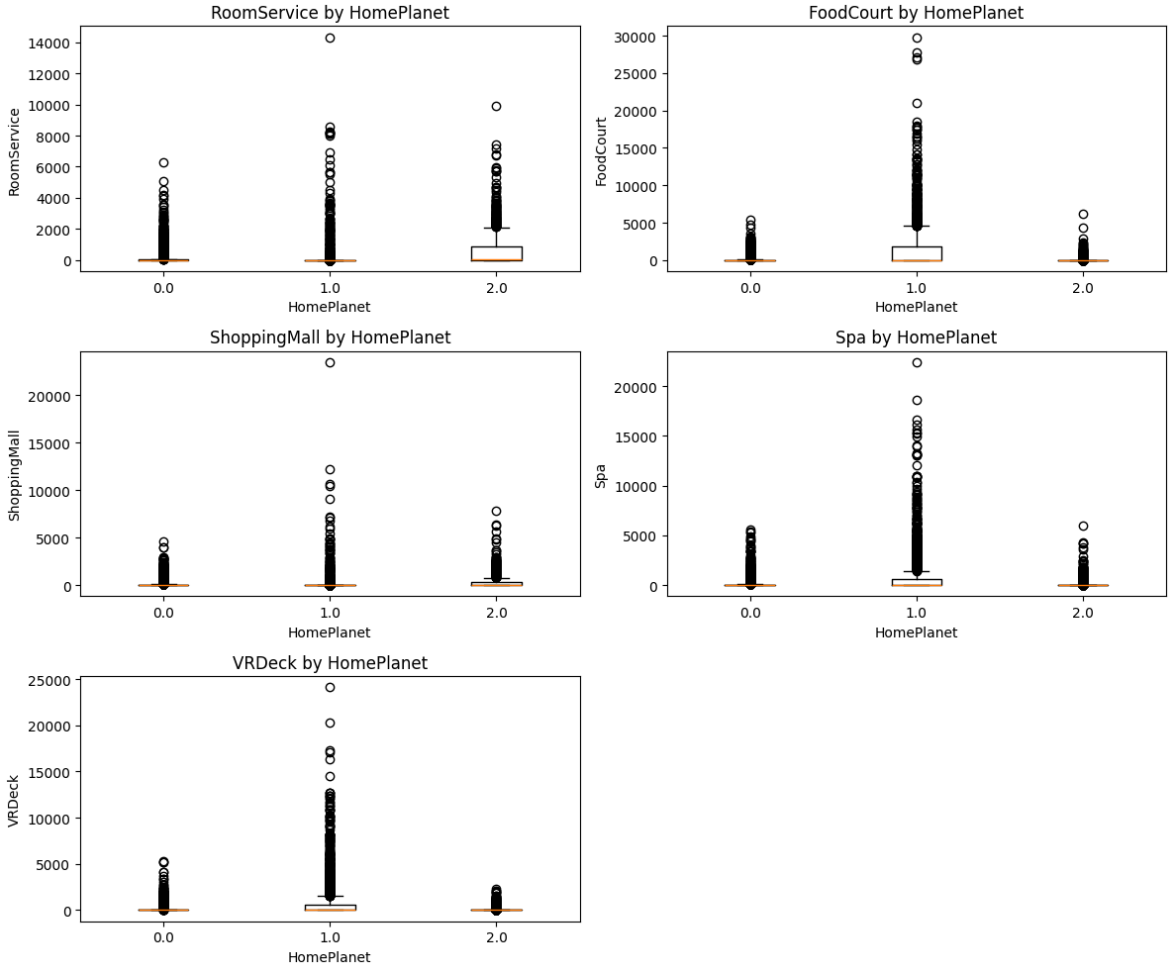
Out[60]:

HomePlanet	RoomService_count	RoomService_mean	RoomService_median	FoodCourt_
0.0	4499	136.94	0.0	
1.0	2095	145.23	0.0	
2.0	1723	552.90	25.0	

In [61]:

```
plot_box_subplots(spaceship_data, money_cols, "HomePlanet", nrows=3, ncols=2, fi
```

```
C:\Users\maxoo\AppData\Local\Temp\ipykernel_8224\1461217584.py:11: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  ax.boxplot(data, labels=labels, showfliers=showfliers)
C:\Users\maxoo\AppData\Local\Temp\ipykernel_8224\1461217584.py:11: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  ax.boxplot(data, labels=labels, showfliers=showfliers)
C:\Users\maxoo\AppData\Local\Temp\ipykernel_8224\1461217584.py:11: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  ax.boxplot(data, labels=labels, showfliers=showfliers)
C:\Users\maxoo\AppData\Local\Temp\ipykernel_8224\1461217584.py:11: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  ax.boxplot(data, labels=labels, showfliers=showfliers)
C:\Users\maxoo\AppData\Local\Temp\ipykernel_8224\1461217584.py:11: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  ax.boxplot(data, labels=labels, showfliers=showfliers)
```



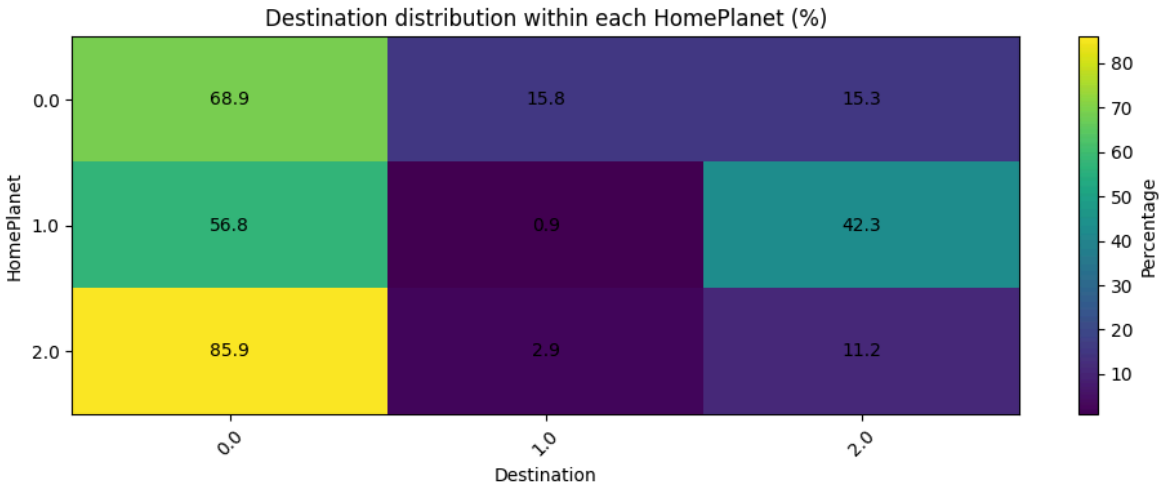
בדיקת יעד לכל כוכב אם באחוזים

In [62]:
```
dest_table, dest_values, dest_rows, dest_cols = crosstab_percent(
    spaceship_data, "HomePlanet", "Destination"
)

dest_table
```

Out[62]:

Destination	0.0	1.0	2.0
HomePlanet			
0.0	68.87	15.81	15.32
1.0	56.78	0.91	42.31
2.0	85.91	2.85	11.24

In [63]:
```
plot_heatmap(
    dest_values,
    dest_rows,
    dest_cols,
    title="Destination distribution within each HomePlanet (%)",
    xlabel="Destination",
    ylabel="HomePlanet"
)
```



הוספת כוכב אם 2 לאנשים שהיעד שלהם הוא כוכב 0 בסבירות של 86 אחוז

In [64]:
```
mask = (spaceship_data["HomePlanet"].isna()) & (spaceship_data["Destination"] ==
```

In [65]:
```
spaceship_data.loc[mask, "HomePlanet"] = 2
```

In [66]:
```
missing_table(spaceship_data, "HomePlanet")
```

Out[66]:

	Count
HomePlanet is NaN	51
HomePlanet not NaN	8642
Total rows	8693

בדיקת שכיחת טווח גילאים מכל כוכב אם

In [67]:
```
# יצירת AgeBin על הדאטה המקורי
bins = [0,10,20,30,40,50,60,70,80,200]
labels = ["0-9","10-19","20-29","30-39","40-49","50-59","60-69","70-79","80+"]

spaceship_data['AgeBin'] = pd.cut(
    spaceship_data['Age'],
    bins=bins,
    labels=labels,
    right=False
)
```

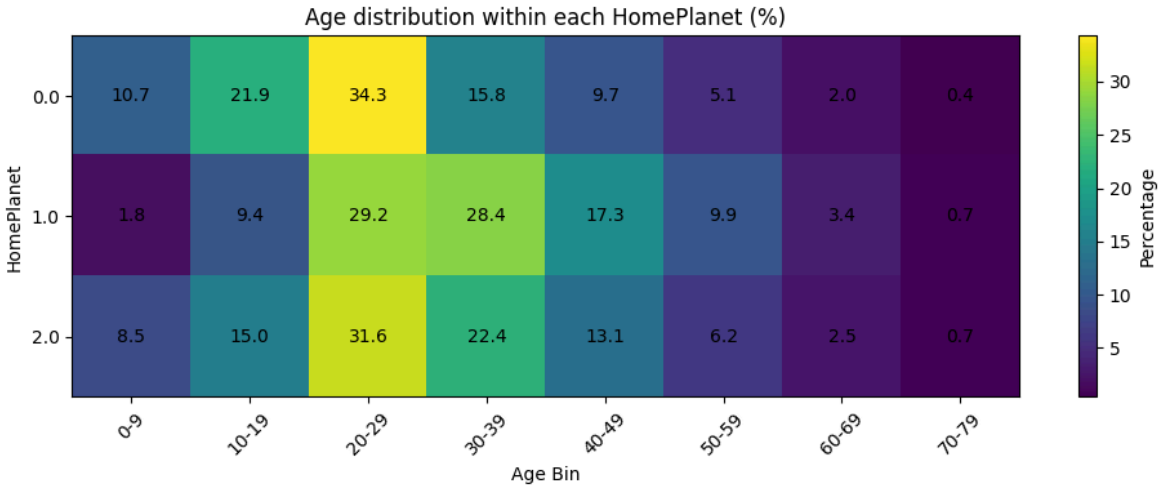
In [68]:
```
# טבלת אחוזים: HomePlanet בתוך AgeBin
age_table, age_values, age_rows, age_cols = crosstab_percent(
    spaceship_data, "HomePlanet", "AgeBin"
)

age_table
```

Out[68]:

AgeBin	0-9	10-19	20-29	30-39	40-49	50-59	60-69	70-79
HomePlanet								
0.0	10.68	21.95	34.32	15.84	9.73	5.07	2.02	0.40
1.0	1.77	9.45	29.21	28.35	17.27	9.88	3.36	0.72
2.0	8.48	14.97	31.65	22.42	13.09	6.17	2.52	0.70

In [69]:

```
# Heatmap
plot_heatmap(
    age_values,
    age_rows,
    age_cols,
    title="Age distribution within each HomePlanet (%)",
    xlabel="Age Bin",
    ylabel="HomePlanet"
)
```



In [70]:

```
spaceship_data = spaceship_data.drop(columns=['AgeBin'])
```

בדיקה כמה אחוז מהוי איי פי הכל כוכב אם

In [71]:

```
vip_homeplanet_pct = category_distribution(
    spaceship_data,
    filter_col="VIP",
    filter_value=1,
    target_col="HomePlanet"
)

vip_homeplanet_pct
```

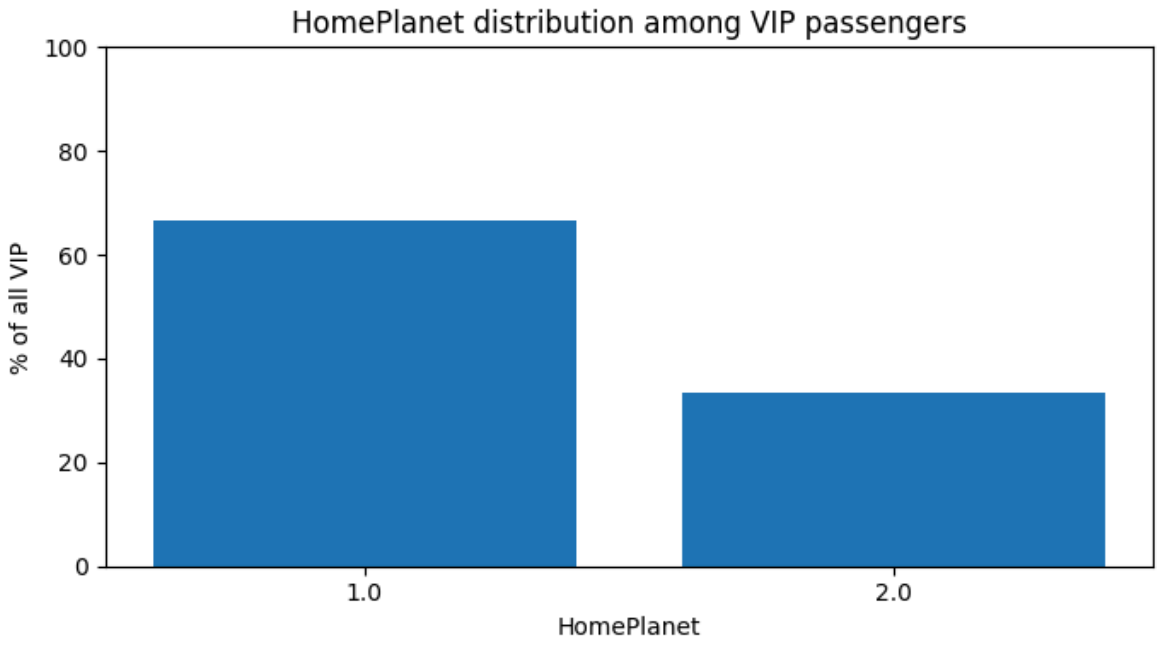
Out[71]:

```
HomePlanet
1.0    66.5
2.0    33.5
Name: proportion, dtype: float64
```

In [72]:

```
plt.figure(figsize=(7, 4))
plt.bar(vip_homeplanet_pct.index.astype(str), vip_homeplanet_pct.values)

plt.ylabel("% of all VIP")
plt.xlabel("HomePlanet")
```

```
plt.title("HomePlanet distribution among VIP passengers")
plt.ylim(0, 100)

plt.tight_layout()
plt.show()
```



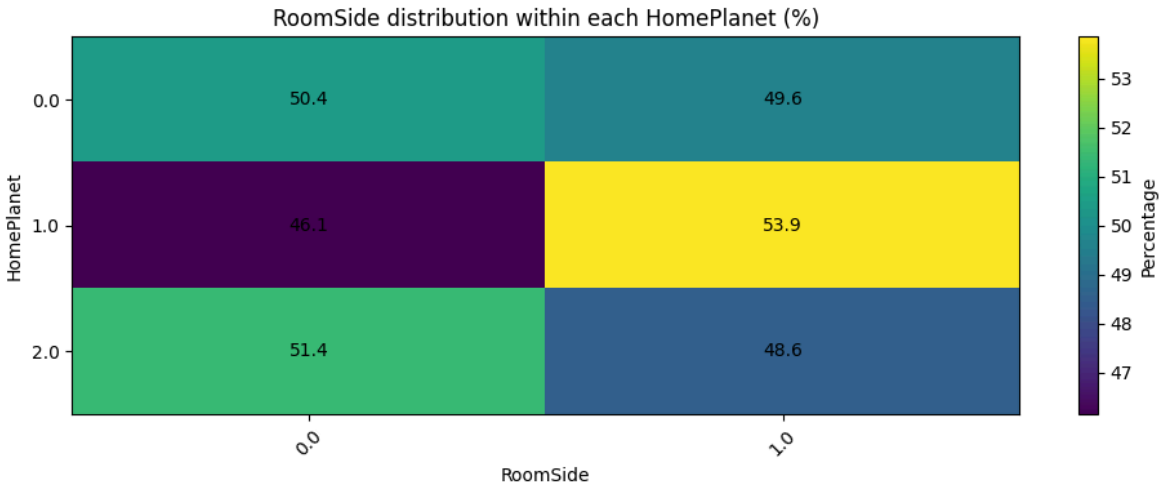
בדיקת צד בספינה

In [73]:
```
side_table, side_values, side_rows, side_cols = crosstab_percent(
    spaceship_data, "HomePlanet", "RoomSide"
)

side_table
```

Out[73]:

RoomSide	0.0	1.0
HomePlanet		
0.0	50.37	49.63
1.0	46.14	53.86
2.0	51.39	48.61

In [74]:
```
plot_heatmap(
    side_values,
    side_rows,
    side_cols,
    title="RoomSide distribution within each HomePlanet (%)",
    xlabel="RoomSide",
    ylabel="HomePlanet"
)
```



homeplanet מציאת קבוצת כל אחד מהנוסעים ללא

In [75]:
```
missing_before = spaceship_data["HomePlanet"].isna().sum()
fill_missing_by_group_unique(spaceship_data, "GroupId", "HomePlanet")
missing_after = spaceship_data["HomePlanet"].isna().sum()

filled_values = missing_before - missing_after

summary_df = pd.DataFrame({
    "Metric": [
        "Missing before filling",
        "Values filled",
        "Missing after filling",
        "Fill rate (%)"
    ],
    "Value": [
        missing_before,
        filled_values,
        missing_after,
        round(100 * filled_values / missing_before, 2)
    ]
})

display(summary_df)
```

	Metric	Value
0	Missing before filling	51.00
1	Values filled	19.00
2	Missing after filling	32.00
3	Fill rate (%)	37.25

בדיקת קשר ביו קומה בספינה לבין כוכב אם

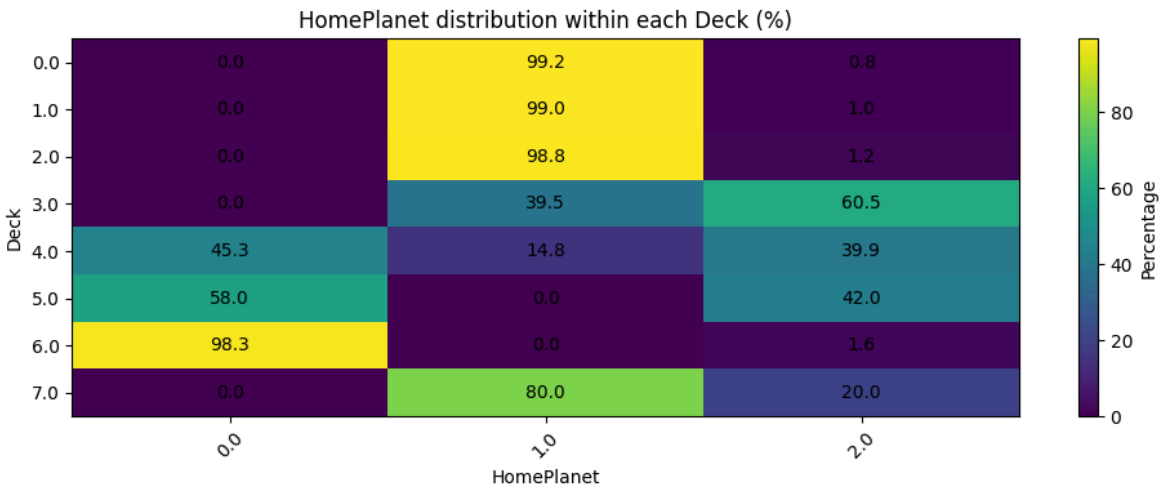
In [76]:
```
deck_hp_table, deck_hp_values, deck_hp_rows, deck_hp_cols = crosstab_percent(
    spaceship_data, "Deck", "HomePlanet"
)

deck_hp_table
```

Out[76]:

HomePlanet	0.0	1.0	2.0
Deck			
0.0	0.00	99.22	0.78
1.0	0.00	98.97	1.03
2.0	0.00	98.79	1.21
3.0	0.00	39.50	60.50
4.0	45.30	14.79	39.91
5.0	57.95	0.00	42.05
6.0	98.35	0.00	1.65
7.0	0.00	80.00	20.00

In [77]:

```
plot_heatmap(
    deck_hp_values,
    deck_hp_rows,
    deck_hp_cols,
    title="HomePlanet distribution within each Deck (%)",
    xlabel="HomePlanet",
    ylabel="Deck"
)
```



Earth זה תמיד G1 Europa מ תמיד גרים אנשים Deck A,B,C,T ב

In [78]:

```
missing_before = spaceship_data['HomePlanet'].isna().sum()
```

In [79]:

```
mask_europa = (
    spaceship_data['HomePlanet'].isna() &
    spaceship_data['Deck'].isin([0, 1, 2, 7])
)
spaceship_data.loc[mask_europa, 'HomePlanet'] = 1  # Europa

mask_earth = (
    spaceship_data['HomePlanet'].isna() &
    (spaceship_data['Deck'] == 6)
)
spaceship_data.loc[mask_earth, 'HomePlanet'] = 0  # Earth
```

In [80]:
```
missing_table(spaceship_data, "HomePlanet")
```

Out[80]:

	Count
HomePlanet is NaN	15
HomePlanet not NaN	8678
Total rows	8693

CryoSleep

In [81]:
```
missing_table(spaceship_data, "CryoSleep")
```

Out[81]:

	Count
CryoSleep is NaN	217
CryoSleep not NaN	8476
Total rows	8693

בדיקת קשר בין שינה ליתר העמודות

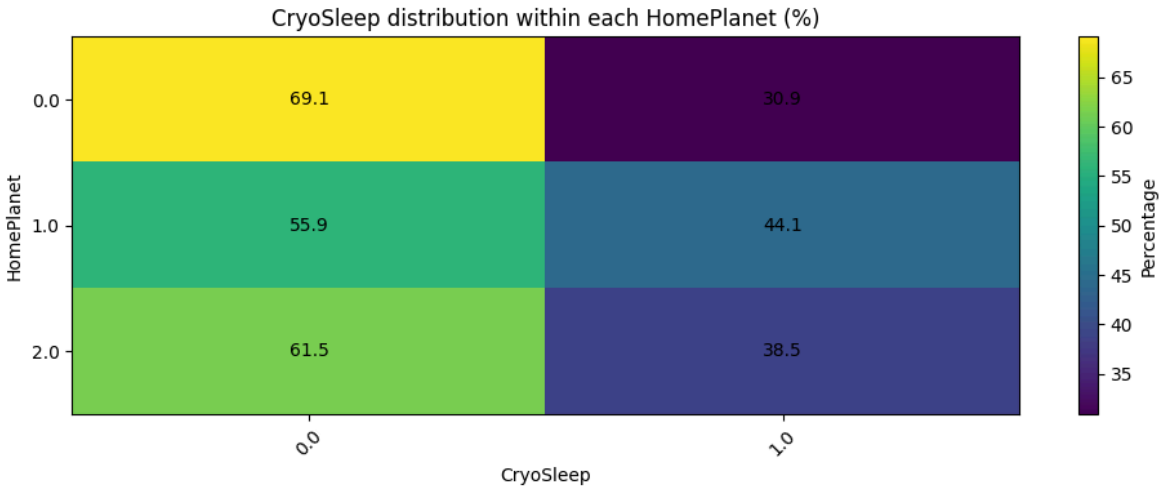
קשר עם כוכב אם

In [82]:
```
# טבלת אחוזים: CryoSleep בתוך HomePlanet
cs_table, cs_values, cs_rows, cs_cols = crosstab_percent(
    spaceship_data, "HomePlanet", "CryoSleep"
)

cs_table
```

Out[82]:

CryoSleep	0.0	1.0
HomePlanet		
0.0	69.12	30.88
1.0	55.90	44.10
2.0	61.49	38.51

In [83]:
```
# Heatmap
plot_heatmap(
    cs_values,
    cs_rows,
    cs_cols,
    title="CryoSleep distribution within each HomePlanet (%)",
    xlabel="CryoSleep",
    ylabel="HomePlanet"
)
```



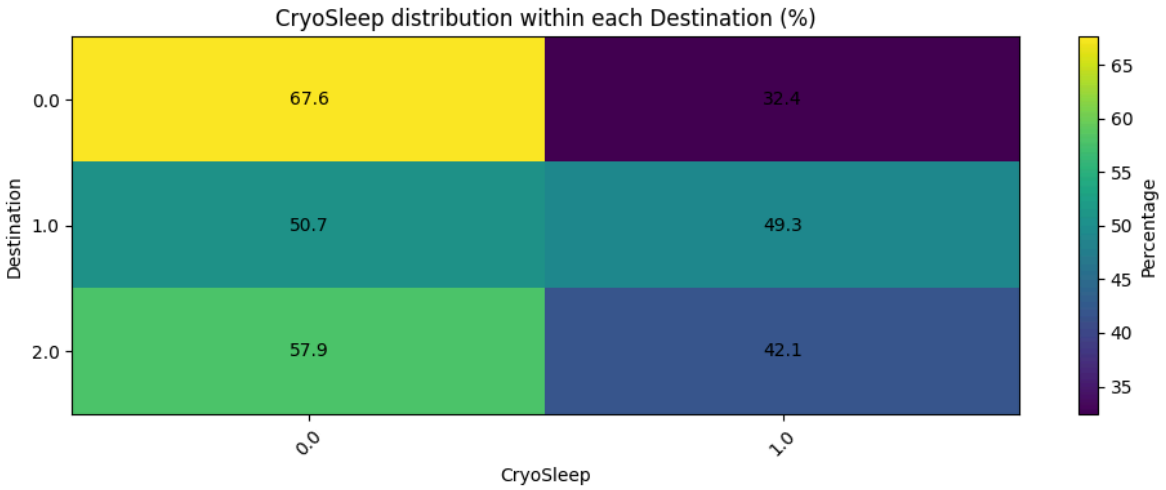
קשר עם יעד

In [84]:
```
# טבלת אחוזים: CryoSleep בתוך Destination
cs_dest_table, cs_dest_values, cs_dest_rows, cs_dest_cols = crosstab_percent(
    spaceship_data, "Destination", "CryoSleep"
)

cs_dest_table
```

Out[84]:

CryoSleep	0.0	1.0
Destination		
0.0	67.61	32.39
1.0	50.71	49.29
2.0	57.92	42.08

In [85]:
```
# Heatmap
plot_heatmap(
    cs_dest_values,
    cs_dest_rows,
    cs_dest_cols,
    title="CryoSleep distribution within each Destination (%)",
    xlabel="CryoSleep",
    ylabel="Destination"
)
```



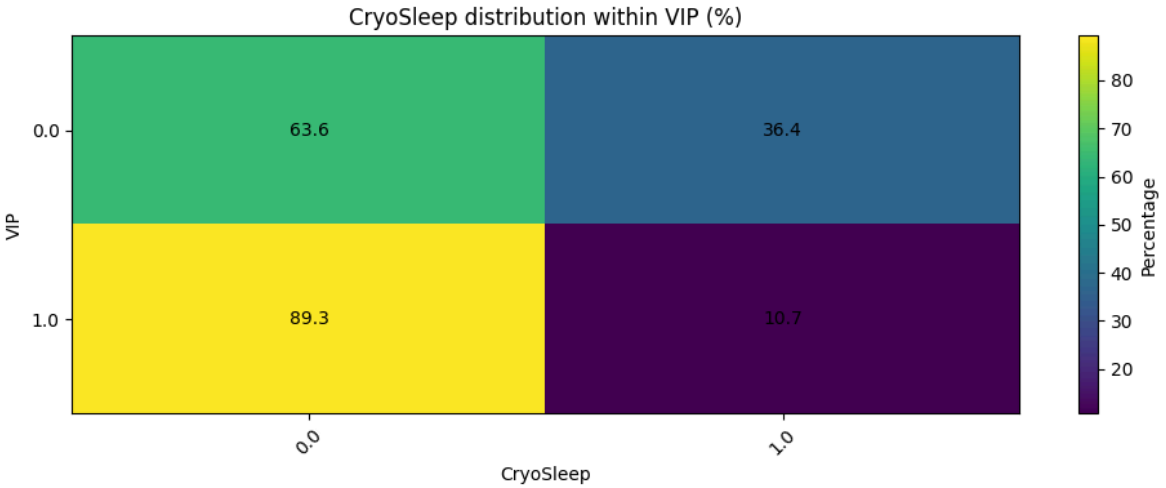
קשר עם וי איי פי

In [86]:
```
# טבלת אחוזים: CryoSleep בתוך VIP
vip_cs_table, vip_cs_values, vip_cs_rows, vip_cs_cols = crosstab_percent(
    spaceship_data, "VIP", "CryoSleep"
)

vip_cs_table
```

Out[86]:

CryoSleep	0.0	1.0
VIP		
0.0	63.62	36.38
1.0	89.29	10.71

In [87]:
```
# Heatmap
plot_heatmap(
    vip_cs_values,
    vip_cs_rows,
    vip_cs_cols,
    title="CryoSleep distribution within VIP (%)",
    xlabel="CryoSleep",
    ylabel="VIP"
)
```



אחוז מהוי איי פי לא ישנים ולכן נמלא 89

In [88]:
```
mask = (
    spaceship_data["CryoSleep"].isna() &
    (spaceship_data["VIP"] == 1)
)

print("Rows to be filled:", mask.sum())
```

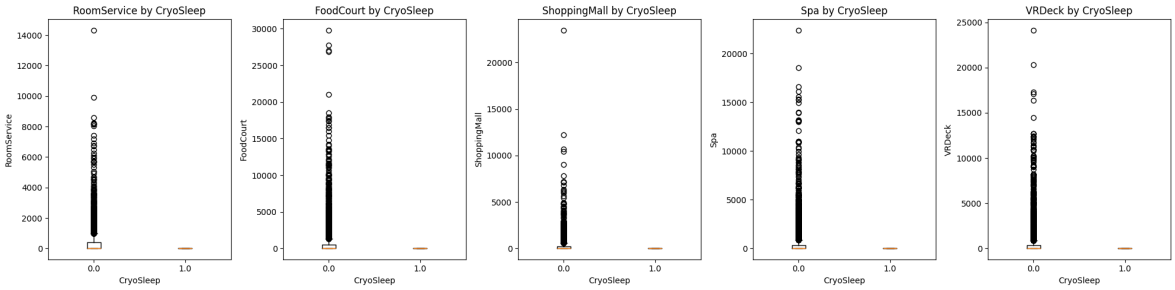
```
Rows to be filled: 3
```

In [89]:
```
spaceship_data.loc[mask, "CryoSleep"] = 0
```

קשר עם הוצאות

In [90]:
```
plot_box_subplots(spaceship_data, money_cols, "CryoSleep", nrows=1, ncols=5, fig
```

```
C:\Users\maxoo\AppData\Local\Temp\ipykernel_8224\1461217584.py:11: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  ax.boxplot(data, labels=labels, showfliers=showfliers)
C:\Users\maxoo\AppData\Local\Temp\ipykernel_8224\1461217584.py:11: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  ax.boxplot(data, labels=labels, showfliers=showfliers)
C:\Users\maxoo\AppData\Local\Temp\ipykernel_8224\1461217584.py:11: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  ax.boxplot(data, labels=labels, showfliers=showfliers)
C:\Users\maxoo\AppData\Local\Temp\ipykernel_8224\1461217584.py:11: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  ax.boxplot(data, labels=labels, showfliers=showfliers)
C:\Users\maxoo\AppData\Local\Temp\ipykernel_8224\1461217584.py:11: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  ax.boxplot(data, labels=labels, showfliers=showfliers)
```



In [91]:
```
spenders_mask = (
    spaceship_data['CryoSleep'].isna() &
    (spaceship_data['TotalSpend'] > 0)
)

filled_count = spenders_mask.sum()

spaceship_data.loc[spenders_mask, 'CryoSleep'] = 0

print(f"Waking up money spenders: {filled_count}")
```

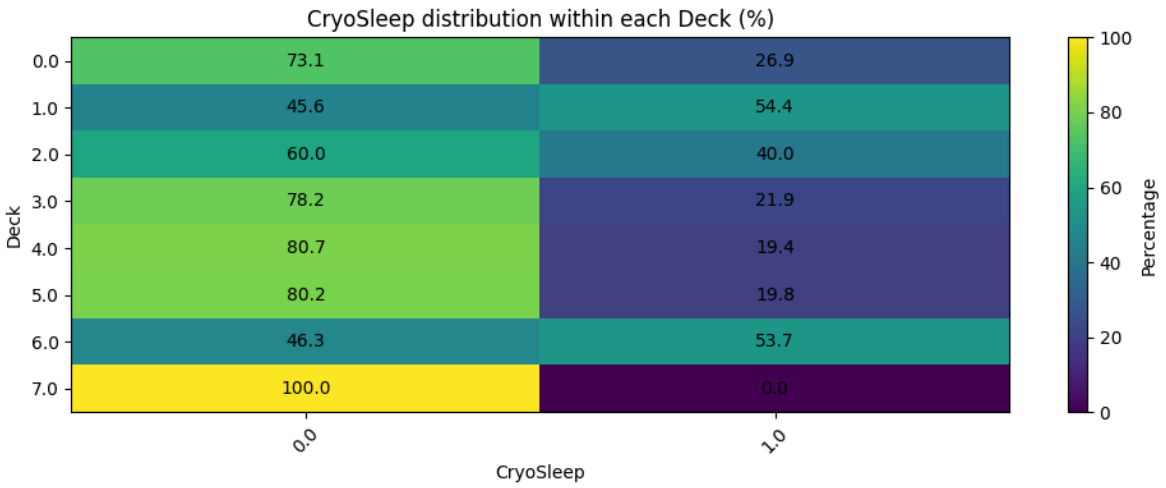
```
Waking up money spenders: 116
```

In [92]:
```
deck_cs_table, deck_cs_values, deck_cs_rows, deck_cs_cols = crosstab_percent(
    spaceship_data, "Deck", "CryoSleep"
)

deck_cs_table
```

Out[92]:

CryoSleep	0.0	1.0
Deck		
0.0	73.12	26.88
1.0	45.58	54.42
2.0	60.00	40.00
3.0	78.15	21.85
4.0	80.65	19.35
5.0	80.18	19.82
6.0	46.34	53.66
7.0	100.00	0.00

In [93]:
```
# Heatmap
plot_heatmap(
    deck_cs_values,
    deck_cs_rows,
    deck_cs_cols,
    title="CryoSleep distribution within each Deck (%)",
    xlabel="CryoSleep",
    ylabel="Deck"
)
```



הכנסת קומות לשינה: T F E D

In [94]:
```
mask_fill_0 = spaceship_data['CryoSleep'].isna() & spaceship_data['Deck'].isin([
count_fill_0 = mask_fill_0.sum()

spaceship_data.loc[mask_fill_0, 'CryoSleep'] = 0

print(f"Filled CryoSleep=0 by Deck (A/D/E/F/T): {count_fill_0}")
```

```
Filled CryoSleep=0 by Deck (A/D/E/F/T): 32
```

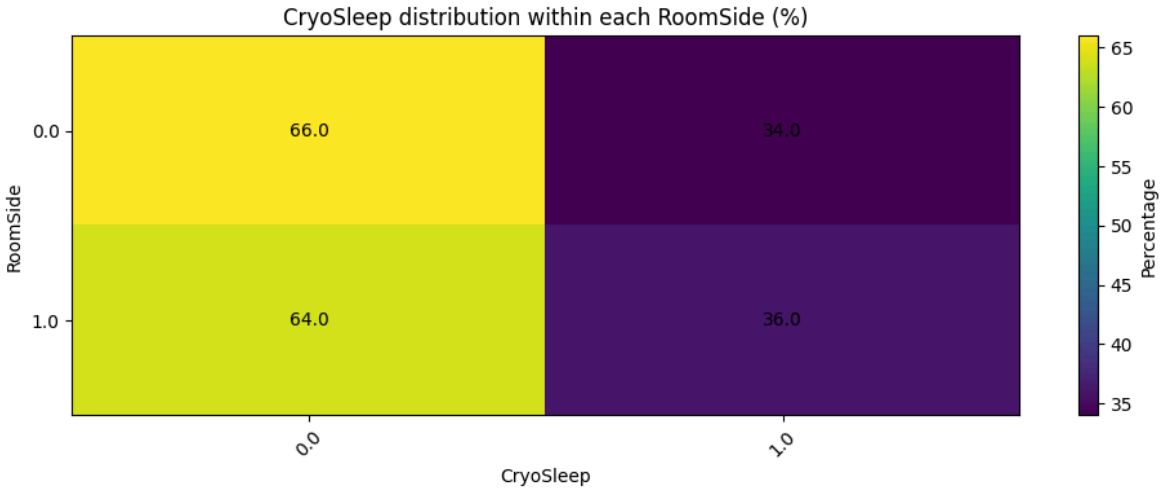
בדיקת צד חדר

In [95]:
```
# טבלת אחוזים: CryoSleep בתוך RoomSide
rs_cs_table, rs_cs_values, rs_cs_rows, rs_cs_cols = crosstab_percent(
    spaceship_data, "RoomSide", "CryoSleep"
)

rs_cs_table
```

Out[95]:

CryoSleep	0.0	1.0
RoomSide		
0.0	66.00	34.00
1.0	63.96	36.04

In [96]:
```
# Heatmap
plot_heatmap(
    rs_cs_values,
    rs_cs_rows,
    rs_cs_cols,
    title="CryoSleep distribution within each RoomSide (%)",
    xlabel="CryoSleep",
    ylabel="RoomSide"
)
```



בדיקת לפי קבוצה

In [97]:
```
def all_neighbors_sleep(x):
    # Alone or all NaN - false
    if x.count() == 0:
        return False
    # 1 neighbour not sleeps (0) - false
    if (x == 0).any():
        return False
    # all known neighbours sleep (1) - true
    return True
```

קבוצת ישנים

In [ ]:
```
# Group check - if all other group members (with known CryoSleep) sleep, fill as
group_sleeps = spaceship_data.groupby('GroupId')['CryoSleep'].transform(all_neig
```

קבינת ישנים

In [ ]:
```
# Cabin check (where not NaN) - if all other cabin members (with known CryoSleep
if 'Cabin' in spaceship_data.columns:
    cabin_sleeps = spaceship_data.groupby('Cabin')['CryoSleep'].transform(all_ne
else:
    cabin_sleeps = pd.Series(False, index=spaceship_data.index)
```

כל הקנדידתים אפשריים

In [100…
```
# CryoSleep empty + spendings 0 + whole group OR whole cabin sleeps = candidates
candidates = (
    spaceship_data['CryoSleep'].isna() &
    (spaceship_data['TotalSpend'] == 0) &
    (group_sleeps | cabin_sleeps)
)

spaceship_data.loc[candidates, 'CryoSleep'] = 1

print(f"Changed to sleeps: {int(candidates.sum())}")

remaining_cryo = int(spaceship_data['CryoSleep'].isna().sum())
print(f"NaNs in CryoSleep: {remaining_cryo}")
```

```
Changed to sleeps: 9
NaNs in CryoSleep: 57
```

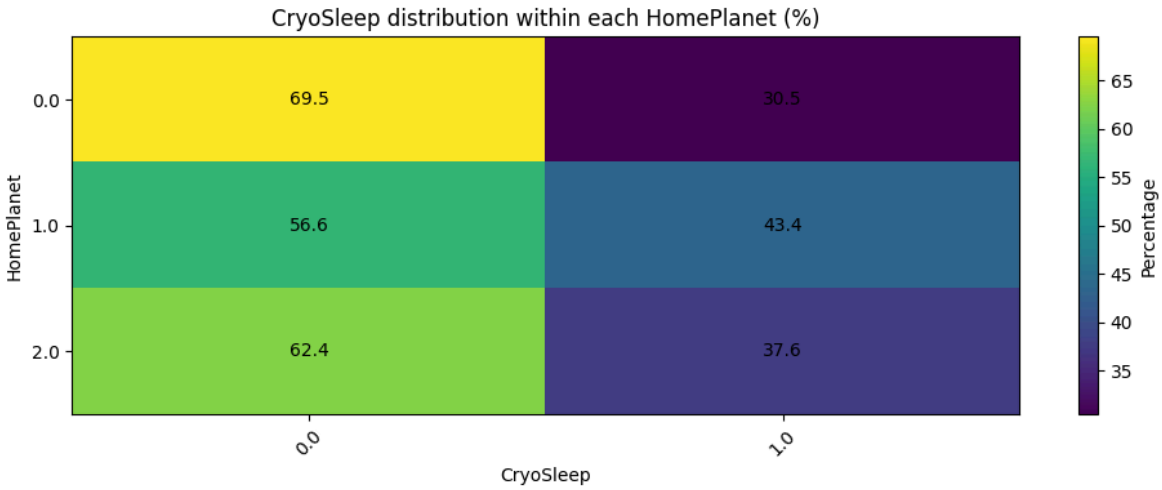
בדיקה לפי יעד סופי

In [101…
```
# טבלת אחוזים: CryoSleep בתוך HomePlanet
cs_hp_table, cs_hp_values, cs_hp_rows, cs_hp_cols = crosstab_percent(
    spaceship_data, "HomePlanet", "CryoSleep"
)

cs_hp_table
```

Out[101…

CryoSleep	0.0	1.0
HomePlanet		
0.0	69.50	30.50
1.0	56.59	43.41
2.0	62.38	37.62

In [102…
```
# Heatmap
plot_heatmap(
    cs_hp_values,
    cs_hp_rows,
    cs_hp_cols,
    title="CryoSleep distribution within each HomePlanet (%)",
    xlabel="CryoSleep",
    ylabel="HomePlanet"
)
```



מילוי 0 בכוכב אם 0

In [103…

```
mask = (
    spaceship_data['CryoSleep'].isna() &
    (spaceship_data['HomePlanet'] == 0)
)

filled_count = int(mask.sum())

spaceship_data.loc[mask, 'CryoSleep'] = 0

print(f"Filled CryoSleep=0 for {filled_count} passengers from HomePlanet=0")
```

```
Filled CryoSleep=0 for 38 passengers from HomePlanet=0
```

מילוי שמי שלא בזבז כסף ישן במהלך הטיסה

In [104…

```
spenders_mask = (
    spaceship_data['CryoSleep'].isna() &
    (spaceship_data['TotalSpend'] == 0)
)

filled_count = spenders_mask.sum()

spaceship_data.loc[spenders_mask, 'CryoSleep'] = 1

print(f"Sleeping money savers: {filled_count}")
```

```
Sleeping money savers: 19
```

כמה אנשים נשארו

In [105…

```
missing_table(spaceship_data, "CryoSleep")
```

Out[105…

	Count
CryoSleep is NaN	0
CryoSleep not NaN	8693
Total rows	8693

Cabin

בדיקה כמה חסר

In [106…
```
missing_table(spaceship_data, "Cabin")
```

Out[106…

	Count
Cabin is NaN	199
Cabin not NaN	8494
Total rows	8693

הוספת ערך לCabin לפי GROUPID

In [107…
```
group_stats = (
    spaceship_data
    .groupby('GroupId')['Cabin']
    .agg(
        unique_cabins=lambda x: x.dropna().nunique(),
        missing_cabins_count=lambda x: x.isna().sum()
    )
)

group_stats[group_stats['missing_cabins_count'] > 0].value_counts()
```

Out[107…
```
unique_cabins  missing_cabins_count
0              1                      99
1              1                      87
2              1                      10
3              1                       3
Name: count, dtype: int64
```

In [108…
```
target_group_ids = group_stats[
    (group_stats['unique_cabins'] == 1) &
    (group_stats['missing_cabins_count'] > 0)
].index

pd.DataFrame({"Num_groups_to_fill": [len(target_group_ids)]})
```

Out[108…

	Num_groups_to_fill
0	87

In [109…
```
fillable_rows = (
    spaceship_data[spaceship_data['GroupId'].isin(target_group_ids)]
    .sort_values(['GroupId'])
)

fillable_rows[['GroupId', 'Cabin']].head(4)
```

Out[109…

	GroupId	Cabin
103	110	NaN
104	110	B/5/P
105	110	B/5/P
106	110	B/5/P

In [110…

```
mask = spaceship_data['GroupId'].isin(target_group_ids)

spaceship_data.loc[mask, 'Cabin'] = (
    spaceship_data.loc[mask]
    .groupby('GroupId')['Cabin']
    .transform(lambda x: x.ffill().bfill())
)
```

פיצול חוזר לערכי התא

In [111…

```
spaceship_data['Deck'] = spaceship_data['Cabin'].str.split('/').str[0]
spaceship_data['RoomNum'] = spaceship_data['Cabin'].str.split('/').str[1].astype
spaceship_data['RoomSide'] = spaceship_data['Cabin'].str.split('/').str[2]
```

בדיקה סופית כמה חסר

In [112…

```
missing_table(spaceship_data, "Cabin")
```

Out[112…

	Count
Cabin is NaN	112
Cabin not NaN	8581
Total rows	8693

Destination

destination_map = { 'TRAPPIST-1e': 0, 'PSO J318.5-22': 1, '55 Cancri e': 2 }

בדיקה כמה חסר

In [113…

```
missing_table(spaceship_data, "Destination")
```

Out[113…

	Count
Destination is NaN	182
Destination not NaN	8511
Total rows	8693

הצגת קשרים בין יתר העמודות

בדיקת קשר לכוכב אם

In [114…

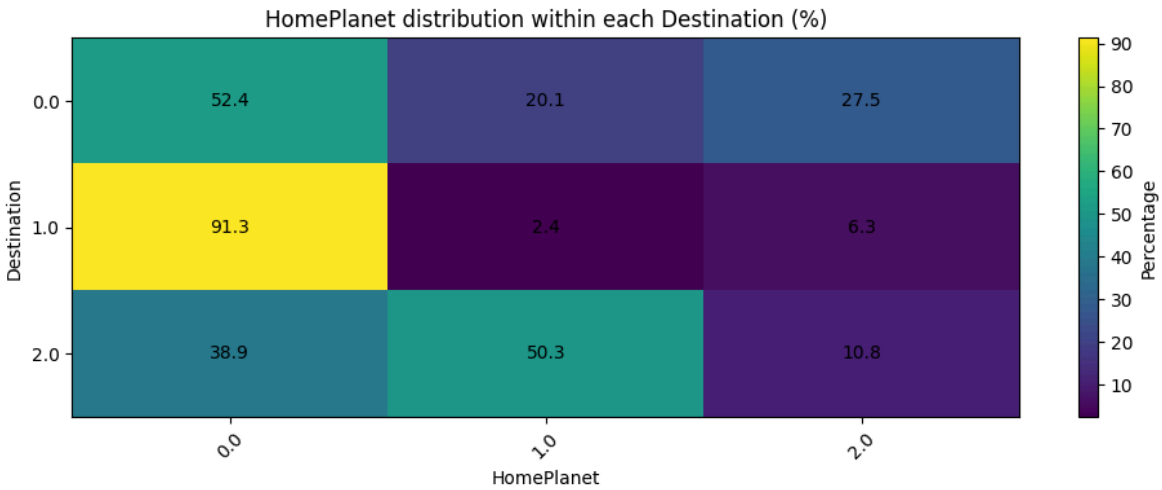
```
# טבלת אחוזים: HomePlanet בתוך Destination
hp_dest_table, hp_dest_values, hp_dest_rows, hp_dest_cols = crosstab_percent(
    spaceship_data, "Destination", "HomePlanet"
)

hp_dest_table
```

Out[114…

HomePlanet	0.0	1.0	2.0
Destination			
0.0	52.43	20.10	27.47
1.0	91.29	2.40	6.31
2.0	38.88	50.28	10.84

In [115…

```
# Heatmap
plot_heatmap(
    hp_dest_values,
    hp_dest_rows,
    hp_dest_cols,
    title="HomePlanet distribution within each Destination (%)",
    xlabel="HomePlanet",
    ylabel="Destination"
)
```



אחוז שכוכב אם 0 מוביל ליעד 1 91

In [116…

```
mask = (
    spaceship_data["HomePlanet"].isna() &
    (spaceship_data["Destination"] == 1)
)

filled_count = int(mask.sum())
spaceship_data.loc[mask, "HomePlanet"] = 0

print(f"Filled HomePlanet=0 for {filled_count} passengers with Destination=1")
```

```
Filled HomePlanet=0 for 4 passengers with Destination=1
```

קשר עם שינה

In [117…

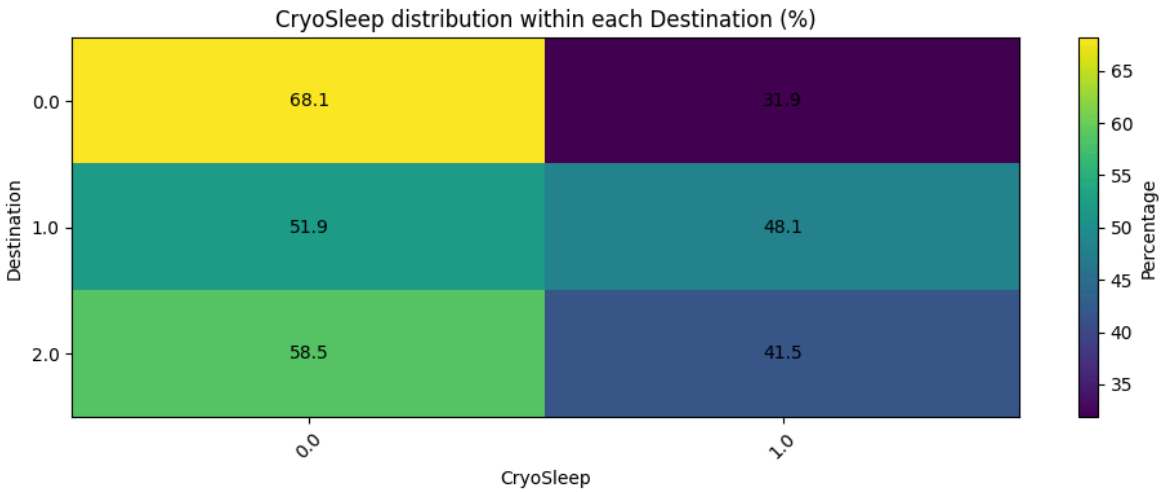
```
# טבלת אחוזים: CryoSleep בתוך Destination
cs_dest_table, cs_dest_values, cs_dest_rows, cs_dest_cols = crosstab_percent(
    spaceship_data, "Destination", "CryoSleep"
)

cs_dest_table
```

Out[117…

CryoSleep	0.0	1.0
Destination		
0.0	68.13	31.87
1.0	51.88	48.12
2.0	58.50	41.50

In [118…

```
# Heatmap
plot_heatmap(
    cs_dest_values,
    cs_dest_rows,
    cs_dest_cols,
    title="CryoSleep distribution within each Destination (%)",
    xlabel="CryoSleep",
    ylabel="Destination"
)
```



קשר עם VIP

In [119…

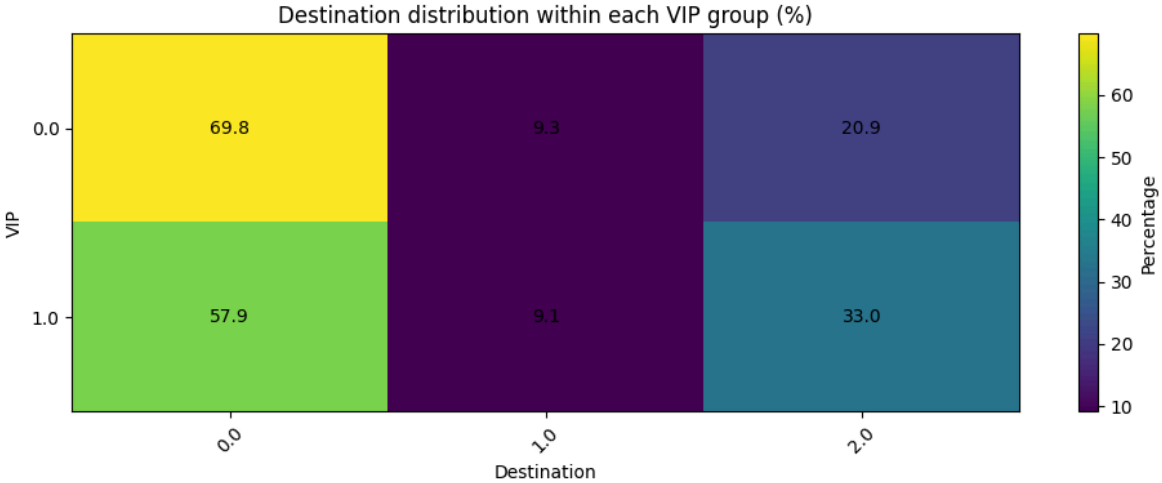
```
# טבלת אחוזים: Destination בתוך VIP
dest_vip_table, dest_vip_values, dest_vip_rows, dest_vip_cols = crosstab_percent
    spaceship_data, "VIP", "Destination"
)

dest_vip_table
```

Out[119…

Destination	0.0	1.0	2.0
VIP			
0.0	69.83	9.32	20.85
1.0	57.87	9.14	32.99

In [120…

```
# Heatmap
plot_heatmap(
    dest_vip_values,
    dest_vip_rows,
    dest_vip_cols,
    title="Destination distribution within each VIP group (%)",
    xlabel="Destination",
    ylabel="VIP"
)
```



בדיקת כמות יעדים שיש לכל קבוצה

In [121…

```
dest_unique_per_group = (
    spaceship_data
    .dropna(subset=['Destination'])
    .groupby('GroupId')['Destination']
    .nunique()
)

dest_unique_per_group.value_counts().sort_index().to_frame('Num_groups')
```

Out[121…

Destination	Num_groups
1	5397
2	668
3	49

יצירת מילון לזיהוי יעדים לפי קבוצה

In [122…

```
fill_missing_by_group_unique(spaceship_data, "GroupId", "Destination")
```

הוספת יעד לנוסעים שלקבוצתם יעד משותף

In [123…
```
missing_table(spaceship_data, "Destination")
```

Out[123…

	Count
Destination is NaN	128
Destination not NaN	8565
Total rows	8693

מציאת יעד לפי נתוני הקבינה

מציאת שורות עם קבינה לא ריקה

In [124…
```
deck_dest_table, deck_dest_values, deck_dest_rows, deck_dest_cols = crosstab_per
    spaceship_data.dropna(subset=["Cabin"]),
    "Deck",
    "Destination"
)

deck_dest_table
```

Out[124…

Destination	0.0	1.0	2.0
Deck			
A	55.38	0.77	43.85
B	55.67	0.38	43.95
C	55.20	1.87	42.93
D	72.50	3.54	23.96
E	81.59	4.83	13.58
F	76.53	8.95	14.52
G	67.24	18.19	14.57
T	100.00	0.00	0.00

ליעד 0 T מילוי קומה

In [125…
```
spaceship_data.loc[
    (spaceship_data["Deck"] == "T") &
    (spaceship_data["Destination"].isna()),
    "Destination"
] = 0
```

In [126…
```
rs_dest_table, rs_dest_values, rs_dest_rows, rs_dest_cols = crosstab_percent(
    spaceship_data.dropna(subset=["Cabin"]),
    "RoomSide",
    "Destination"
)
```

```
rs_dest_table
```

Out[126…

Destination	0.0	1.0	2.0
RoomSide			
P	69.93	9.51	20.56
S	69.20	9.13	21.67

בדיקת חדרים לפי יעד שלהם

In [127…

```
dest_per_cabin = (
    spaceship_data
    .dropna(subset=['Cabin'])
    .groupby('Cabin')['Destination']
    .nunique()
)

dest_per_cabin.value_counts().sort_index().to_frame("Num_cabins")
```

Out[127…

	Num_cabins
Destination	
0	103
1	5832
2	591
3	34

יצירת מילון מקבינה לאן היעד שלה במקרה שכל חברי הקבינה לאותו המקום

In [128…

```
cabin_destination_map = (
    spaceship_data
    .dropna(subset=['Cabin', 'Destination'])
    .groupby('Cabin')['Destination']
    .agg(lambda s: s.iloc[0] if s.nunique() == 1 else np.nan)
)
```

מילוי היעדים לכל מי שאפשר

In [129…

```
mask = spaceship_data['Destination'].isna() & spaceship_data['Cabin'].notna()

spaceship_data.loc[mask, 'Destination'] = (
    spaceship_data.loc[mask, 'Cabin'].map(cabin_destination_map)
)
```

In [130…

```
print("Missing Destination after cabin fill:", int(spaceship_data['Destination']
```

```
Missing Destination after cabin fill: 124
```

Age

In [131…
```
missing_table(spaceship_data, "Age")
```

Out[131…

	Count
Age is NaN	179
Age not NaN	8514
Total rows	8693

In [132…
```
spaceship_data['GroupSize'] = spaceship_data.groupby('GroupId')['GroupId'].trans
spaceship_data['IsAlone'] = (spaceship_data['GroupSize'] == 1).astype(int)
```

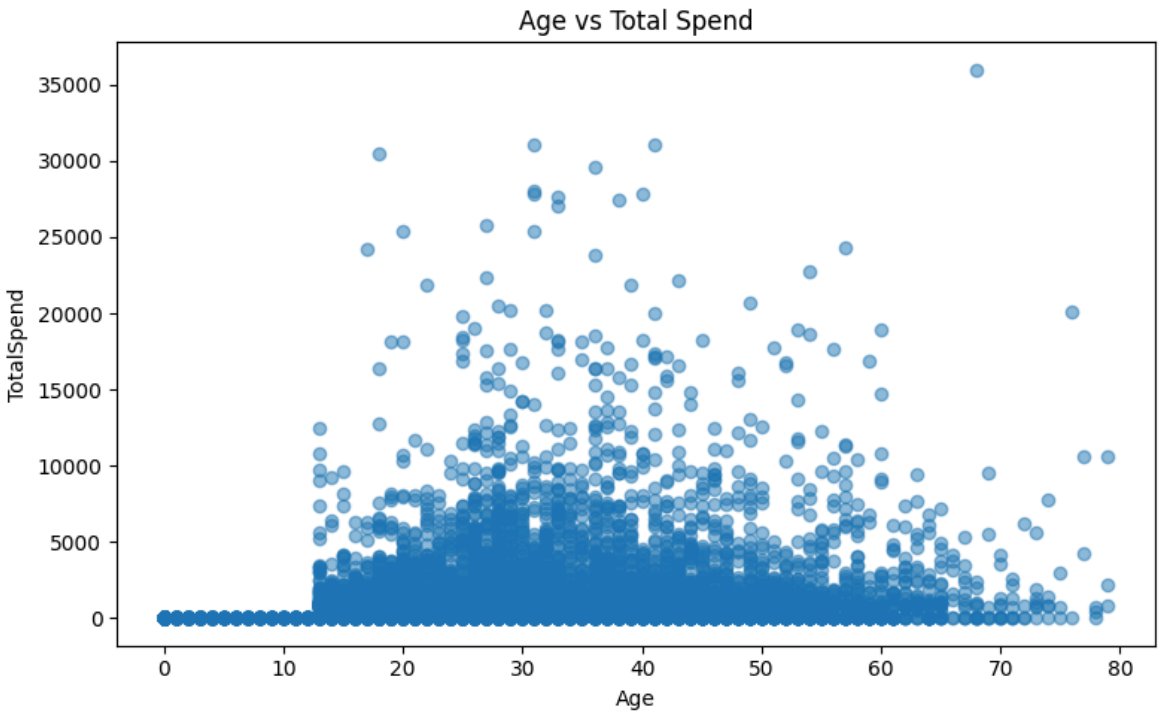
In [133…
```
money_cols = ['RoomService', 'FoodCourt', 'ShoppingMall', 'Spa', 'VRDeck']

spaceship_data['TotalSpend'] = (
    spaceship_data[money_cols]
    .fillna(0)
    .sum(axis=1)
)
```

In [134…
```
mask_ready = spaceship_data['Age'].notna() & spaceship_data['Deck'].notna()
```

In [135…
```
eda_data = spaceship_data.loc[mask_ready, ['Age','GroupId','Deck','GroupSize','I
```

בדיקת הוצאות כספים לפי גיל

In [136…
```
scatter_plot(
    spaceship_data,
    x_col="Age",
    y_col="TotalSpend",
    mask=mask_ready,
    title="Age vs Total Spend"
)
```



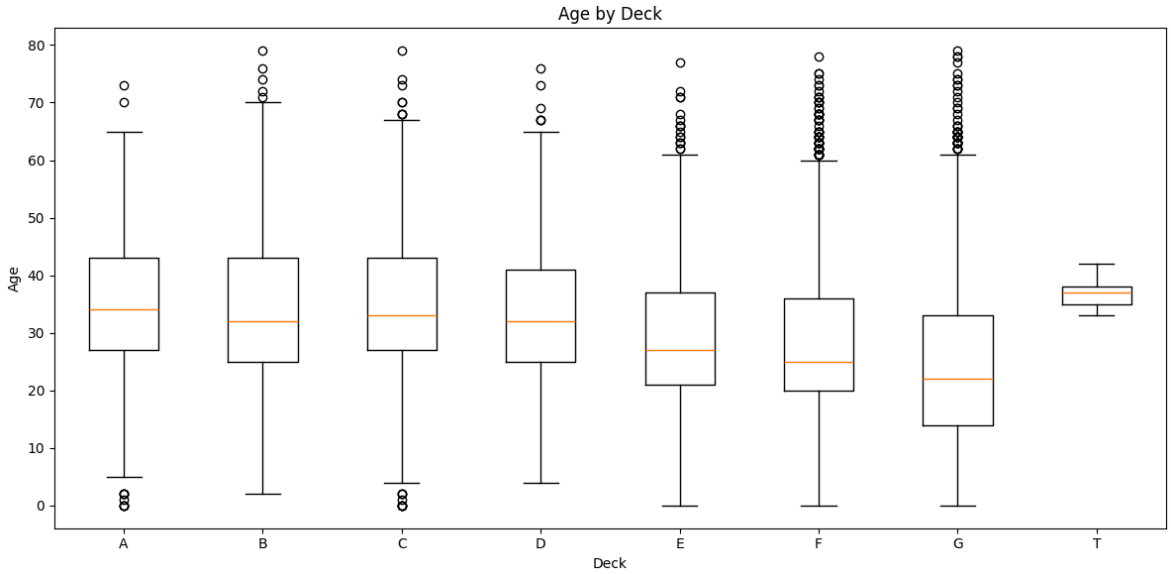
בדיקת קשרים עם יתר העמודות

בדיקה עם קומה

In [137…

```
plot_box_subplots(
    spaceship_data,
    cols=["Age"],
    group_col="Deck",
    nrows=1,
    ncols=1,
    figsize=(12,6)
)
```

```
C:\Users\maxoo\AppData\Local\Temp\ipykernel_8224\1461217584.py:11: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  ax.boxplot(data, labels=labels, showfliers=showfliers)
```

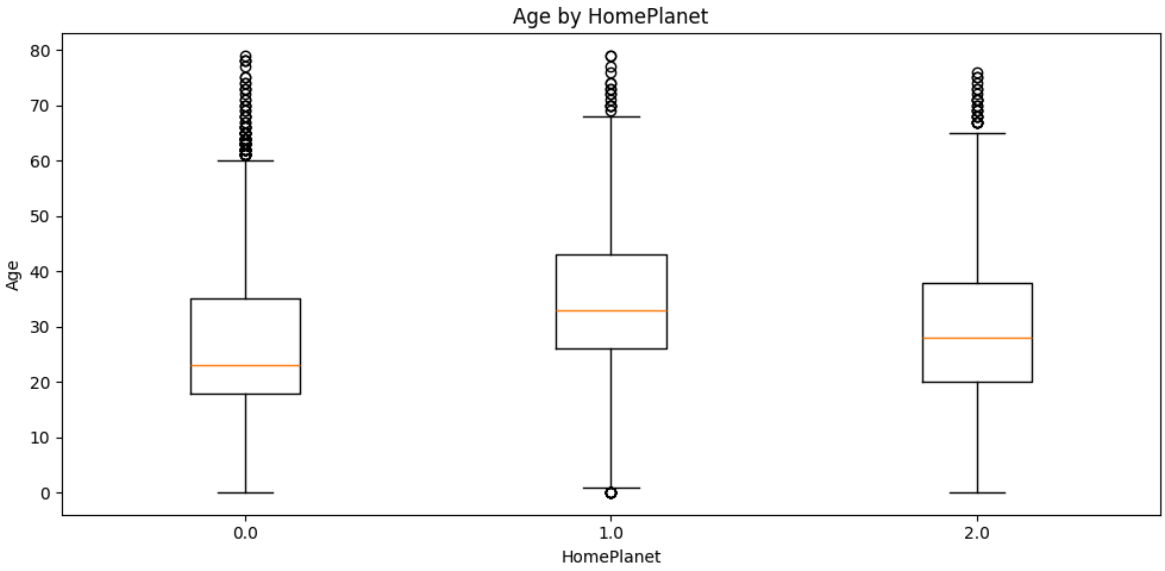


בדיקת עם כוכב אם

In [138…

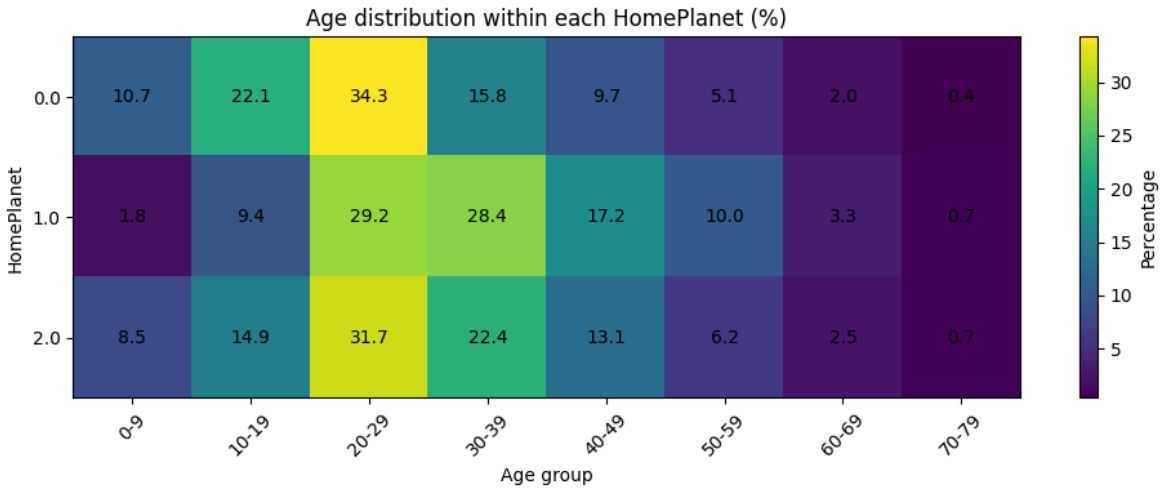
```
plot_box_subplots(
    spaceship_data,
    cols=["Age"],
    group_col="HomePlanet",
    nrows=1,
    ncols=1,
    figsize=(10,5)
)
```

```
C:\Users\maxoo\AppData\Local\Temp\ipykernel_8224\1461217584.py:11: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  ax.boxplot(data, labels=labels, showfliers=showfliers)
```



In [139…

```
bins = [0, 10, 20, 30, 40, 50, 60, 70, 80, 200]
labels = ["0-9", "10-19", "20-29", "30-39", "40-49", "50-59", "60-69", "70-79",

age_bin = pd.cut(spaceship_data["Age"], bins=bins, labels=labels, right=False)

tmp = pd.DataFrame({
    "HomePlanet": spaceship_data["HomePlanet"],
    "AgeBin": age_bin
})

age_hp_table, age_hp_values, age_hp_rows, age_hp_cols = crosstab_percent(tmp, "H
age_hp_table

plot_heatmap(age_hp_values, age_hp_rows, age_hp_cols,
             title="Age distribution within each HomePlanet (%)",
             xlabel="Age group", ylabel="HomePlanet")
```



בדיקה עם יעד

In [140…

```
# AgeBin מחוץ לדאטה
age_bin = pd.cut(
    spaceship_data["Age"],
    bins=bins,
    labels=labels,
    right=False
)
```

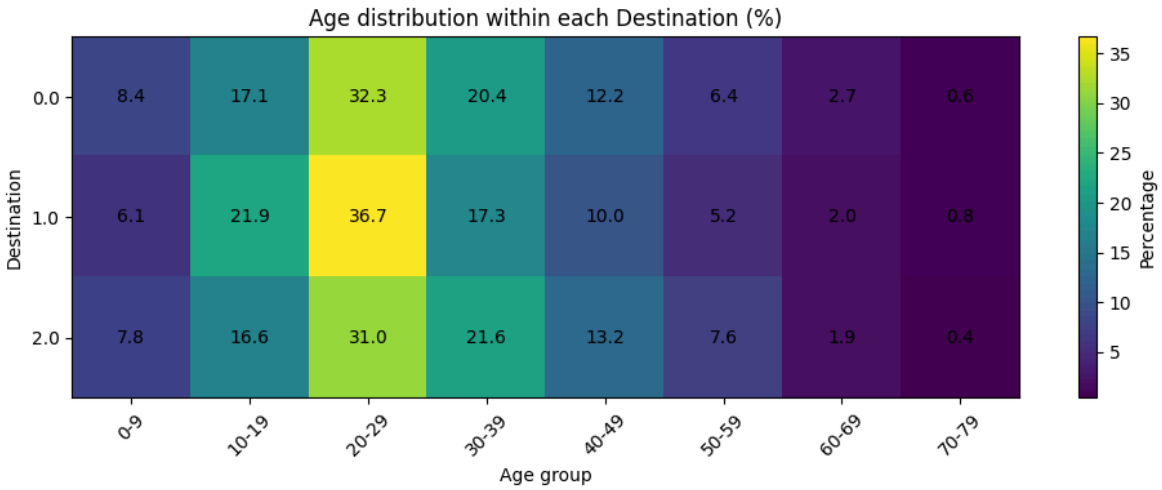
```
tmp = pd.DataFrame({
    "Destination": spaceship_data["Destination"],
    "AgeBin": age_bin
})

# טבלת אחוזים: AgeBin בתוך Destination
age_dest_table, age_dest_values, age_dest_rows, age_dest_cols = crosstab_percent
    tmp, "Destination", "AgeBin"
)

age_dest_table
```

Out[140…

AgeBin	0-9	10-19	20-29	30-39	40-49	50-59	60-69	70-79
Destination								
0.0	8.37	17.05	32.28	20.43	12.24	6.36	2.71	0.57
1.0	6.09	21.95	36.68	17.26	10.03	5.20	2.03	0.76
2.0	7.84	16.59	31.04	21.56	13.15	7.56	1.86	0.40

In [141…

```
# Heatmap
plot_heatmap(
    age_dest_values,
    age_dest_rows,
    age_dest_cols,
    title="Age distribution within each Destination (%)",
    xlabel="Age group",
    ylabel="Destination"
)
```



בדיקה עם שינה

In [142…

```
tmp = pd.DataFrame({
    "AgeBin": age_bin,
    "CryoSleep": spaceship_data["CryoSleep"]
})

cs_age_table, cs_age_values, cs_age_rows, cs_age_cols = crosstab_percent(
    tmp, "AgeBin", "CryoSleep"
)
```

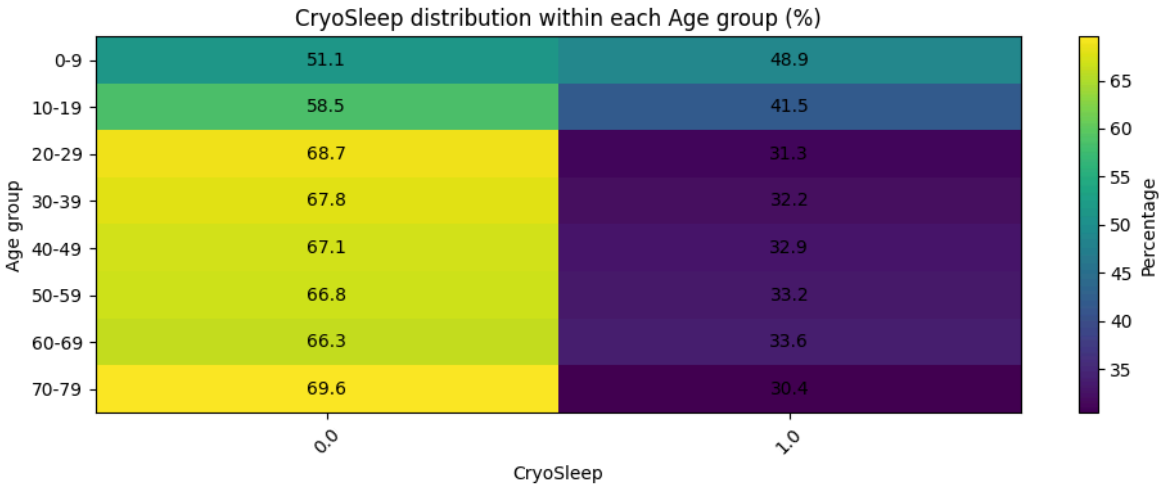
```
cs_age_table
```

Out[142…

CryoSleep	0.0	1.0
AgeBin		
0-9	51.10	48.90
10-19	58.55	41.45
20-29	68.68	31.32
30-39	67.78	32.22
40-49	67.08	32.92
50-59	66.79	33.21
60-69	66.35	33.65
70-79	69.57	30.43

In [143…

```
plot_heatmap(
    cs_age_values,
    cs_age_rows,
    cs_age_cols,
    title="CryoSleep distribution within each Age group (%)",
    xlabel="CryoSleep",
    ylabel="Age group"
)
```



In [144…

```
spaceship_data['Age'] = spaceship_data.groupby('Deck')['Age'].transform(lambda x
```

VIP

חיפוש כמה יי אייי פי חסרים

In [145…

```
vip_missing_table = (
    spaceship_data['VIP']
    .isna()
    .value_counts()
    .rename({True: "Missing", False: "Known"})
    .to_frame("Count")
```

```
)

vip_missing_table
```

Out[145…

VIP	Count
Known	8490
Missing	203

בדיקה בין ארץ מוצא לוי איי פי

In [146…

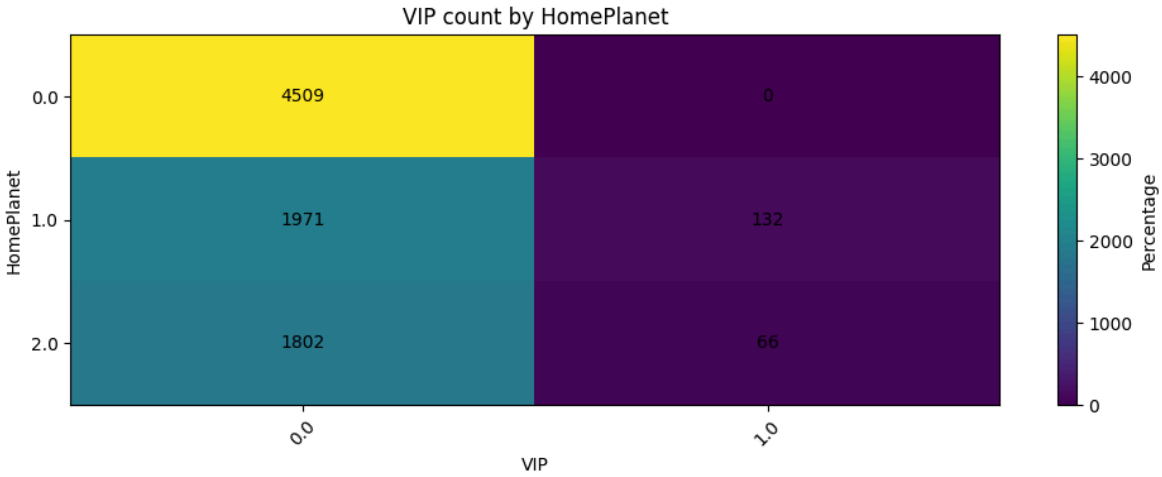
```
tmp = spaceship_data[['HomePlanet', 'VIP']].dropna()

vip_hp_table = pd.crosstab(tmp['HomePlanet'], tmp['VIP'])
vip_hp_table
```

Out[146…

HomePlanet	VIP 0.0	VIP 1.0
0.0	4509	0
1.0	1971	132
2.0	1802	66

In [147…

```
plot_heatmap(
    vip_hp_table.values,
    vip_hp_table.index.astype(str),
    vip_hp_table.columns.astype(str),
    title="VIP count by HomePlanet",
    xlabel="VIP",
    ylabel="HomePlanet",
    fmt="d"
)
```



כל הנוסעים מכדור הארץ אינם וי איי פי

In [148…

```
earth_mask = spaceship_data['VIP'].isna() & (spaceship_data['HomePlanet'] == 0)
```

```
spaceship_data.loc[earth_mask, 'VIP'] = 0
```

In [149…

```
print(f"Filled VIP = 0 for Earth: {earth_mask.sum()}")
```

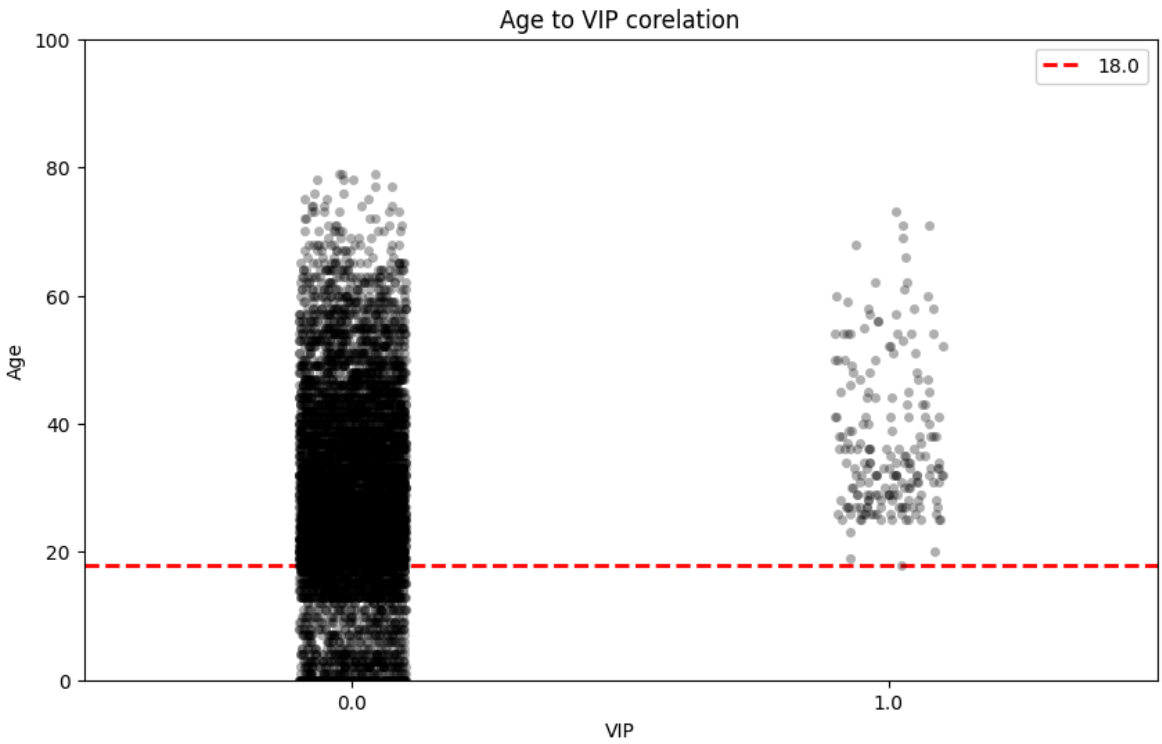
```
Filled VIP = 0 for Earth: 117
```

בדיקה עם וי איי פי

In [150…

```
clean_data = spaceship_data.loc[
    (spaceship_data['Age'] != -1) &
    (spaceship_data['VIP'] != -1)
]

min_vip_age = clean_data[clean_data['VIP'] == 1]['Age'].min()
print(f"The most young VIP: {min_vip_age} years")

plt.figure(figsize=(10, 6))
sns.stripplot(data=clean_data, x='VIP', y='Age', color='black', alpha=0.3, jitte
plt.axhline(y=min_vip_age, color='red', linestyle='--', linewidth=2, label=min_v
plt.title('Age to VIP corelation')
plt.legend()
plt.ylim(0, 100)
plt.show()
```

```
The most young VIP: 18.0 years
```



In [151…

```
kids_mask = (spaceship_data['VIP']== -1) & (spaceship_data['Age'] < min_vip_age)
spaceship_data.loc[kids_mask, 'VIP'] = 0
print(f"VIP=False for kids: {kids_mask.sum()}")
```

```
VIP=False for kids: 0
```

חיבור עם שינה

In [152…

```
tmp = spaceship_data[['CryoSleep', 'VIP']].dropna()
```

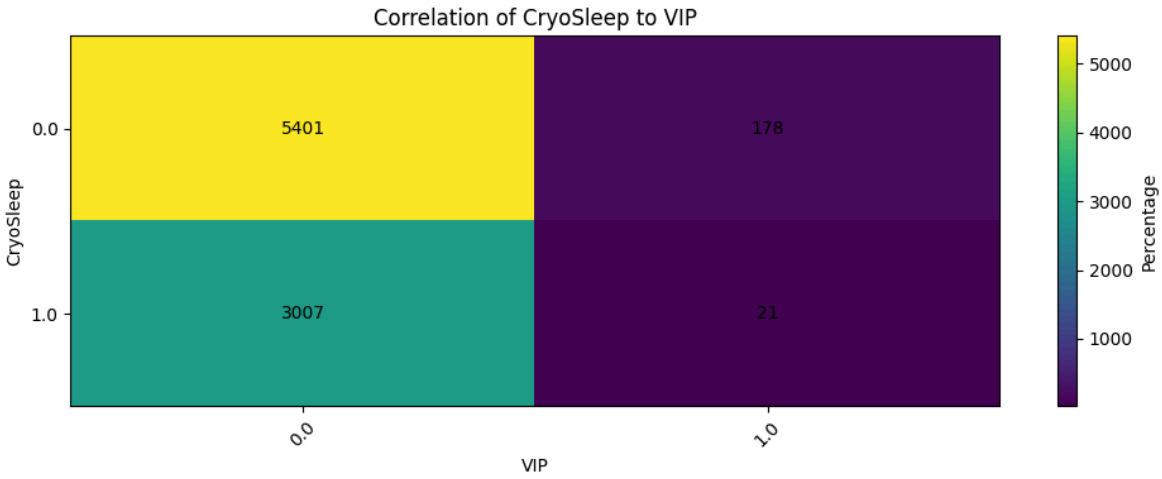
```
cryo_vip_table = pd.crosstab(tmp['CryoSleep'], tmp['VIP'])
cryo_vip_table
```

Out[152…

VIP	0.0	1.0
CryoSleep		
0.0	5401	178
1.0	3007	21

In [153…

```
plot_heatmap(
    cryo_vip_table.values,
    cryo_vip_table.index.astype(str),
    cryo_vip_table.columns.astype(str),
    title="Correlation of CryoSleep to VIP",
    xlabel="VIP",
    ylabel="CryoSleep",
    fmt="d"
)
```



חיבור עם יעד

In [154…

```
tmp = spaceship_data[['Destination', 'VIP']].dropna()

dest_vip_table = pd.crosstab(tmp['Destination'], tmp['VIP'])
dest_vip_table
```

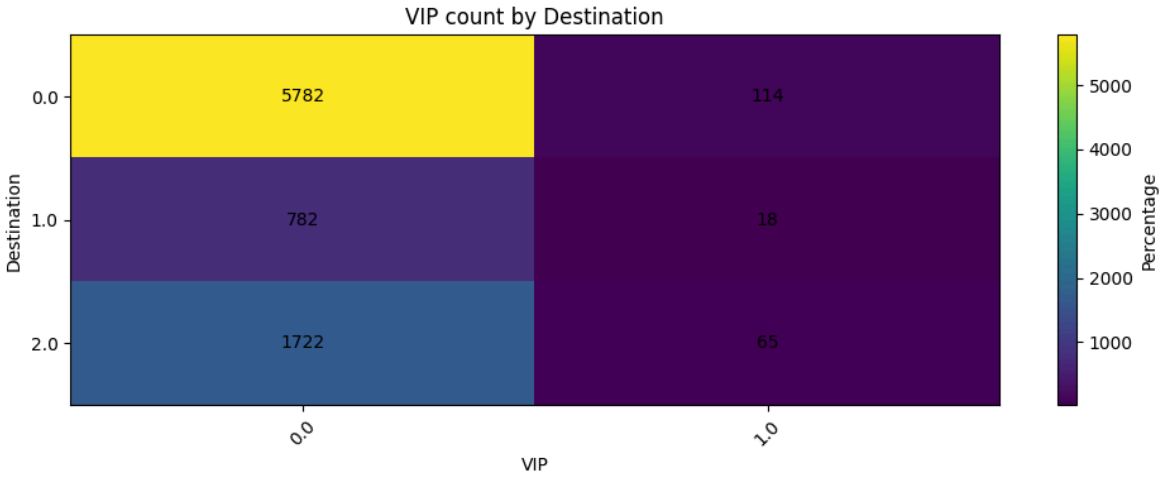
Out[154…

VIP	0.0	1.0
Destination		
0.0	5782	114
1.0	782	18
2.0	1722	65

In [155…

```
plot_heatmap(
    dest_vip_table.values,
    dest_vip_table.index.astype(str),
    dest_vip_table.columns.astype(str),
    title="VIP count by Destination",
    xlabel="VIP",
```

```
    ylabel="Destination",
    fmt="d"
)
```



MONEY

הוספת 0 לכל עמודה שביתר העמודות יש אפסים

In [156…
```
money_cols = ["RoomService", "FoodCourt", "ShoppingMall", "Spa", "VRDeck"]
```

In [157…
```
col_means = {}

for col in money_cols:
    valid = spaceship_data[(spaceship_data[col].notna()) & (spaceship_data[col]
    col_means[col] = valid.mean()
    print(f"{col} mean (excluding 0 and NaN) = {col_means[col]}")
```

```
RoomService mean (excluding 0 and NaN) = 651.6323679727427
FoodCourt mean (excluding 0 and NaN) = 1276.436476751801
ShoppingMall mean (excluding 0 and NaN) = 508.6583850931677
Spa mean (excluding 0 and NaN) = 831.0706214689266
VRDeck mean (excluding 0 and NaN) = 861.3920265780731
```

הוספת ממוצע לכל עמודה לאנשים שמבזבזים כסף (לא כולל את האפסים)

In [158…
```
for col in money_cols:
    mask_nan = spaceship_data[col].isna()
    other_cols = [c for c in money_cols if c != col]

    # (נספר כ-0 רק לבדיקה NaN) חסר, בדוק אם שאר העמודות כולן 0 col לכל השורות שבהן
    other_all_zero = spaceship_data.loc[mask_nan, other_cols].fillna(0).eq(0).al

    # אינדקסים למילוי 0
    idx_zero = spaceship_data.loc[mask_nan].index[other_all_zero]
    spaceship_data.loc[idx_zero, col] = 0

    # אינדקסים למילוי mean
    idx_mean = spaceship_data.loc[mask_nan].index[~other_all_zero]
    spaceship_data.loc[idx_mean, col] = col_means[col]
```

In [159…

```
for col in money_cols:
    mask_nan = spaceship_data[col].isna()
    other_cols = [c for c in money_cols if c != col]

    # לכל השורות שבהן col 0 בדוק אם שאר העמודות כולן ,חסר (NaN רק לבדיקה 0-נספר כ)
    other_all_zero = spaceship_data.loc[mask_nan, other_cols].fillna(0).eq(0).al

    # 0 אינדקסים למילוי
    idx_zero = spaceship_data.loc[mask_nan].index[other_all_zero]
    spaceship_data.loc[idx_zero, col] = 0

    # אינדקסים למילוי mean
    idx_mean = spaceship_data.loc[mask_nan].index[~other_all_zero]
    spaceship_data.loc[idx_mean, col] = col_means[col]
```

In [160…

```
print("Remaining NaNs in money columns:")
print(spaceship_data[money_cols].isna().sum())
```

```
Remaining NaNs in money columns:
RoomService     0
FoodCourt       0
ShoppingMall    0
Spa             0
VRDeck          0
dtype: int64
```

תיקוני דאטה

In [161…

```
spaceship_data.replace(-1, np.nan).isna().sum().to_frame('Num_NaN')
```

Out[161…

	Num_NaN
HomePlanet	11
CryoSleep	0
Cabin	112
Destination	124
Age	112
VIP	86
RoomService	0
FoodCourt	0
ShoppingMall	0
Spa	0
VRDeck	0
Name	200
Transported	0
TotalSpend	0
GroupId	0
GroupMemberId	0
Deck	112
RoomNum	112
RoomSide	112
FirstName	200
Surname	200
GroupSize	0
IsAlone	0

הורדת עמודות לא רלוונטיות

In [162…
```
spaceship_data['TotalSpend'] = spaceship_data[money_cols].sum(axis=1)
```

In [163…
```
final_data = spaceship_data.drop(columns=['Name', 'FirstName', 'Surname', 'Cabin
```

In [164…
```
final_data.replace(-1, np.nan, inplace=True)
```

In [165…
```
final_data.isna().sum().to_frame('Num_NaN')
```

Out[165…

	Num_NaN
HomePlanet	11
CryoSleep	0
Destination	124
Age	112
VIP	86
RoomService	0
FoodCourt	0
ShoppingMall	0
Spa	0
VRDeck	0
Transported	0
TotalSpend	0
GroupId	0
GroupMemberId	0
Deck	112
RoomNum	112
RoomSide	112
GroupSize	0

In [166…

```
deck_map = {
    'A': 0,
    'B': 1,
    'C': 2,
    'D': 3,
    'E': 4,
    'F': 5,
    'G': 6,
    'T': 7
}

roomside_map = {
    'P': 0,
    'S': 1
}
```

In [167…

```
final_data['Deck'] = final_data['Deck'].map(deck_map)
final_data['RoomSide'] = final_data['RoomSide'].map(roomside_map)
```

In [168…

```
final_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8693 entries, 0 to 8692
Data columns (total 18 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   HomePlanet     8682 non-null   float64
 1   CryoSleep      8693 non-null   float64
 2   Destination    8569 non-null   float64
 3   Age            8581 non-null   float64
 4   VIP            8607 non-null   float64
 5   RoomService    8693 non-null   float64
 6   FoodCourt      8693 non-null   float64
 7   ShoppingMall   8693 non-null   float64
 8   Spa            8693 non-null   float64
 9   VRDeck         8693 non-null   float64
 10  Transported    8693 non-null   int64
 11  TotalSpend     8693 non-null   float64
 12  GroupId        8693 non-null   int64
 13  GroupMemberId  8693 non-null   int64
 14  Deck           8581 non-null   float64
 15  RoomNum        8581 non-null   float64
 16  RoomSide       8581 non-null   float64
 17  GroupSize      8693 non-null   int64
dtypes: float64(14), int64(4)
memory usage: 1.2 MB
```

השלמות סופיות

vip כולם ללא הוצאות, כולם ללא

In [169…

```
money_cols = ['TotalSpend']

mask = (final_data['VIP'] == 0) & (final_data['CryoSleep'] == 0)

all_zero = (
    (final_data.loc[mask, money_cols].fillna(0) == 0)
    .all(axis=1)
)

percent_all_zero = all_zero.mean() * 100

print(f"Out of non-VIP passengers who were NOT in CryoSleep:")
print(f"{percent_all_zero:.2f}% spent 0 in ALL money categories")
```

```
Out of non-VIP passengers who were NOT in CryoSleep:
10.63% spent 0 in ALL money categories
```

מחיקת השורות הנותרות שלא מצאנו להם שום קשר

In [170…

```
final_data = final_data.dropna(subset=['CryoSleep'])
```

---

homeplanet

מחיקת שורות שלא נמצא קשרים אליהם

In [171…
```
final_data = final_data.dropna(subset=['HomePlanet'])
```

Destination

In [172…
```
dest = final_data[final_data['Destination'].isna()]
print(f"Final NaNs in Destination: {final_data['Destination'].isna().sum()}")
```

```
Final NaNs in Destination: 123
```

מחיקת שורות ללא destination

In [173…
```
final_data = final_data.dropna(subset=['Destination'])
```

In [174…
```
final_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 8559 entries, 0 to 8692
Data columns (total 18 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   HomePlanet     8559 non-null   float64
 1   CryoSleep      8559 non-null   float64
 2   Destination    8559 non-null   float64
 3   Age            8452 non-null   float64
 4   VIP            8474 non-null   float64
 5   RoomService    8559 non-null   float64
 6   FoodCourt      8559 non-null   float64
 7   ShoppingMall   8559 non-null   float64
 8   Spa            8559 non-null   float64
 9   VRDeck         8559 non-null   float64
 10  Transported    8559 non-null   int64
 11  TotalSpend     8559 non-null   float64
 12  GroupId        8559 non-null   int64
 13  GroupMemberId  8559 non-null   int64
 14  Deck           8452 non-null   float64
 15  RoomNum        8452 non-null   float64
 16  RoomSide       8452 non-null   float64
 17  GroupSize      8559 non-null   int64
dtypes: float64(14), int64(4)
memory usage: 1.2 MB
```

VIP

מילוי כל הנותרים כלא VIP

In [175…
```
final_data.replace(np.nan, 0, inplace=True)
```

Age

מילוי גיל ממוצע בכל קומה

In [176…
```
final_data['Age'] = (
    final_data
    .groupby('Deck')['Age']
```

```
    .transform(lambda x: x.fillna(x.median()))
)
```

One_hot Encoding

In [177…

```
# יצירת עמודות one-hot
homeplanet_dummies = pd.get_dummies(
    final_data["HomePlanet"],
    prefix="HomePlanet"
)

destination_dummies = pd.get_dummies(
    final_data["Destination"],
    prefix="Destination"
)

# חיבור לעמודות המקוריות
final_data = pd.concat(
    [final_data, homeplanet_dummies, destination_dummies],
    axis=1
)

# מחיקת העמודות המקוריות (אם אתה רוצה)
final_data.drop(
    ["HomePlanet", "Destination"],
    axis=1,
    inplace=True
)
```

In [ ]:

בדיקה סופית ושמירת הדאטה

In [178…

```
final_data.isna().sum().to_frame('Num_NaN')
```

Out[178…

	Num_NaN
CryoSleep	0
Age	0
VIP	0
RoomService	0
FoodCourt	0
ShoppingMall	0
Spa	0
VRDeck	0
Transported	0
TotalSpend	0
GroupId	0
GroupMemberId	0
Deck	0
RoomNum	0
RoomSide	0
GroupSize	0
HomePlanet_0.0	0
HomePlanet_1.0	0
HomePlanet_2.0	0
Destination_0.0	0
Destination_1.0	0
Destination_2.0	0

In [179…

```
final_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 8559 entries, 0 to 8692
Data columns (total 22 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CryoSleep        8559 non-null   float64
 1   Age              8559 non-null   float64
 2   VIP              8559 non-null   float64
 3   RoomService      8559 non-null   float64
 4   FoodCourt        8559 non-null   float64
 5   ShoppingMall     8559 non-null   float64
 6   Spa              8559 non-null   float64
 7   VRDeck           8559 non-null   float64
 8   Transported      8559 non-null   int64
 9   TotalSpend       8559 non-null   float64
 10  GroupId          8559 non-null   int64
 11  GroupMemberId    8559 non-null   int64
 12  Deck             8559 non-null   float64
 13  RoomNum          8559 non-null   float64
 14  RoomSide         8559 non-null   float64
 15  GroupSize        8559 non-null   int64
 16  HomePlanet_0.0   8559 non-null   bool
 17  HomePlanet_1.0   8559 non-null   bool
 18  HomePlanet_2.0   8559 non-null   bool
 19  Destination_0.0  8559 non-null   bool
 20  Destination_1.0  8559 non-null   bool
 21  Destination_2.0  8559 non-null   bool
dtypes: bool(6), float64(12), int64(4)
memory usage: 1.2 MB
```

In [180…

```
final_data.to_csv("../final_data.csv", index=False)
```